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Conformal Uncertainty for Operator Deep Smoothing of Implied-Volatility Surfaces

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Abstract

Graph Neural Operators (GNOs) can construct smooth implied volatility surfaces from irregular option quotes in a single forward pass, but a deterministic GNO does not quantify uncertainty around its estimate. This thesis develops a post-hoc conformal prediction framework that adds calibrated lower and upper surfaces to a fixed, pre-trained GNO without modifying or retraining the model. Two economically motivated nonconformity scores are compared: a spread-normalised price residual (Score A) and a Vega-weighted implied volatility residual (Score B). Calibration uses the five most recent market surfaces, while a 4×4 Mondrian partition assigns spatially varying quantiles across maturity and normalised moneyness. The bands are subsequently adjusted together to enforce a calendar condition and a total-variance convexity condition based on neighbouring grid points, while keeping implied volatility positive and ensuring that the lower bound remains below the upper bound.

The empirical study uses 46 retained S&P 500 option surfaces collected between December 2025 and August 2026. The first 15 surfaces form the initial calibration period and the remaining 31 are evaluated in walk-forward order. On each surface, 80% of the quotes are supplied to the GNO as context and 20% are withheld for calibration or evaluation. With a nominal coverage level of 90%, Score B achieves mean post-projection coverage of 90.84% under global calibration and 92.86% under Mondrian calibration, compared with 85.46% and 84.27%, respectively, for Score A. The mean band widths are 0.027187 implied-volatility units for Score B and 0.067806 for Score A, making the Score A bands approximately 2.49 times wider. The projection reduces all violations of the two enforced grid conditions to zero for both scores and produces no crossings at the evaluated target quotes. Total-variance convexity is a shape regularisation and does not establish butterfly no-arbitrage in option prices.

Within this dataset, Score B provides the stronger combination of empirical coverage, sharpness, and stability across surfaces. The results show that conformal prediction can serve as a practical uncertainty layer for neural-operator volatility smoothing. Because the observations are temporally ordered, the reported coverage is an empirical walk-forward result rather than a claim of the standard exchangeable-data guarantee.

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Chapter 1

Introduction

Options markets generate large and rapidly changing collections of quotes across strikes and maturities. Cboe Global Markets reported that 4.6 billion contracts were traded across its four options exchanges in 2025, the sixth consecutive annual record. S&P 500 Index (SPX) options accounted for 970.6 million contracts, with an average daily volume of 3.9 million [7]. At this scale, constructing implied volatility (IV) surfaces efficiently from irregular market data is practically important for pricing and risk measurement.

Implied volatility is the volatility parameter that reproduces a quoted option price in the Black-Scholes model. Collecting these values across strikes and maturities gives an implied volatility surface for a particular market snapshot. The surface is not observed continuously: market quotes are available only at a discrete and changing set of locations and may contain noise or temporary inconsistencies. A usable estimate must therefore smooth the observations while retaining the financial structure required of an option-price surface [10].

Traditional parametric approaches, including the Stochastic Volatility Inspired (SVI) model [9], fit a prescribed functional form to each new snapshot. This generally requires a separate numerical optimisation whenever the available quotes change. Gonon, Jacquier, and Wiedemann [10] instead formulate implied volatility smoothing as an operator-learning problem. Their Graph Neural Operator (GNO) accepts observations at irregular locations and produces a smooth surface in one forward pass. It can process changing input configurations without first interpolating every snapshot onto a fixed grid, and its training objective includes penalties for relevant financial shape conditions.

The trained GNO nevertheless returns a deterministic point estimate. In this thesis, uncertainty refers to the discrepancy between that estimate and the market implied volatility at quote locations withheld from the GNO input. The resulting bands describe uncertainty around the reconstructed surface at observed locations; they are not forecasts of future realised volatility or confidence intervals for the GNO parameters. This distinction is important because prediction errors, quote density, liquidity, and option sensitivity differ across the surface.

Monte Carlo dropout and deep ensembles can quantify uncertainty, but they require repeated stochastic evaluations or multiple trained models [8, 11]. Conformal prediction offers a post-hoc alternative that calibrates prediction errors from a fixed model. Under exchangeability of the calibration and test observations, the standard construction provides finite-sample marginal coverage without imposing a parametric error distribution [19, 2]. The data are ordered over time, and quotes from the same market surface are related. Therefore, the exchangeability assumption required for the standard conformal guarantee may not hold. This thesis reports empirical coverage from a walk-forward evaluation instead of claiming a formal coverage guarantee.

Applying conformal prediction in this setting requires a score that reflects the heterogeneous scale of errors across the option domain. This thesis compares a price residual normalised by the bid and ask

spread with an implied volatility residual weighted by Vega. Related research has studied conformal prediction for operator learning in scientific applications [14] and has used implied volatility as a predictor of future realised volatility [6]. The literature reviewed in Chapter 2, however, does not address post-hoc conformal bands around a GNO-smoothed implied volatility surface constructed from irregular option quotes. This is the gap examined in the remainder of the thesis.

1.1 Motivation and Problem Statement

The central problem addressed in this thesis is the construction of calibrated uncertainty bands for implied volatility surfaces produced by the Graph Neural Operator approach. More specifically, this work develops an efficient uncertainty quantification layer that:

1. Targets finite-sample marginal coverage across observed surface points under exchangeability, without requiring a parametric model for the prediction errors;
2. Accounts for heteroskedastic prediction errors through economically motivated normalisation;
3. Responds to changing market conditions through rolling calibration windows;
4. Produces uncertainty bands in both implied volatility and price spaces; and
5. Applies a post-processing routine that enforces a calendar condition and a total-variance convexity condition based on neighbouring grid points on the lower and upper band surfaces.

Options data are heterogeneous across both time and the strike-expiry domain. Liquidity, quote density, bid-ask spreads, and price sensitivity vary between regions of the surface. The proposed framework therefore allows the band width to vary across observed points and evaluates whether spatially adaptive calibration improves coverage across the surface while retaining marginal evaluation over the held-out target quotes.

1.2 Research Objectives

This thesis pursues the following specific objectives:

1. **Design economically motivated nonconformity scores:** Develop two heteroskedastic scoring mechanisms that scale prediction errors using (a) bid-ask spreads in price space and (b) Vega-based weights in implied volatility space, so that uncertainty bands reflect trading frictions and option sensitivity.
2. **Implement rolling window calibration:** Establish a temporal adaptation mechanism using a sliding window of recent market snapshots to compute conformal quantiles, allowing the framework to respond to changes in recently observed prediction errors.
3. **Evaluate spatial heterogeneity via Mondrian conformal prediction:** Investigate whether partitioning the strike-expiry domain into spatial bins and computing region-specific quantiles improves coverage uniformity across areas with varying liquidity.
4. **Construct prediction bands in dual spaces:** Provide practitioners with uncertainty estimates in both implied volatility space (for volatility trading) and price space (for delta hedging and risk management).

5. **Apply calendar and total-variance convexity adjustments to prediction bands:** Apply a projection algorithm to the lower and upper band surfaces and evaluate the calendar and total-variance convexity conditions on the finite (ρ, z) grid used by the implementation.

6. **Validate empirical coverage:** Assess whether the conformal band surfaces achieve the nominal marginal coverage probability $(1 - \alpha)$ across held-out target quote locations in S&P 500 options data spanning December 2025 to August 2026.

1.3 Contributions

The main contributions of this thesis are:

1. **Conformal uncertainty quantification for GNO-based volatility smoothing:** Integration of post-hoc conformal calibration with a fixed, pre-trained Graph Neural Operator to construct lower and upper uncertainty surfaces around implied volatility estimates from irregular option quotes. This work develops and evaluates the integration specifically for implied volatility smoothing, building on earlier conformal prediction research for operator learning in scientific applications [14].

2. **Economically aligned nonconformity scores:** Adaptation of spread-normalised price errors and Vega-weighted implied volatility errors for conformal calibration around the GNO surface. These constructions incorporate bid-ask spread information and option sensitivity when determining the band widths.

3. **Temporal adaptation via rolling calibration:** Use of a rolling window of recent calibration data so that the conformal quantiles respond to changes in the observed prediction errors. Its performance on the temporally ordered surfaces is assessed empirically through the walk-forward evaluation.

4. **Grid-based projection of uncertainty bands:** Implementation of a procedure that applies the calendar and total-variance convexity adjustments used in this thesis to both the lower and upper band surfaces, together with diagnostics for residual violations and lower-upper crossings.

5. **Empirical validation on real market data:** Comprehensive evaluation on S&P 500 options data spanning December 2025 to August 2026, providing evidence that conformal prediction can deliver empirically calibrated uncertainty surfaces in a challenging real-world financial application.

1.4 Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2: Background and Related Work provides the necessary background on implied volatility surfaces, no-arbitrage constraints, traditional smoothing methods, neural operators, and the theoretical foundations of conformal prediction. We review related work on uncertainty quantification in financial machine learning and position our contribution within this literature.

Chapter 3: Methodology presents the technical details of our conformal uncertainty quantification framework. We formally define the problem, introduce the spread-normalised and Vega-weighted nonconformity scores, describe the rolling window calibration mechanism, explain Mondrian conformal prediction for spatial heterogeneity, detail the construction of prediction bands in both price and implied volatility spaces, and present the grid-based projection procedure used to apply calendar and total-variance convexity adjustments to the resulting band surfaces.

Chapter 4: Implementation describes the software architecture, data preprocessing pipeline, and integration with the pre-trained GNO model from Gonon et al.

Chapter 5: Experiments and Results reports the empirical evaluation on S&P 500 options data. We present coverage and sharpness results, visualisations of the uncertainty bands, the comparison of Score A and Score B, the comparison of global and Mondrian calibration, and the projection diagnostics.

Chapter 6: Discussion interprets the results, examines the practical implications for risk management and trading, discusses the limitations of the approach (particularly regarding the exchangeability assumption), and considers the financial validity of the uncertainty bands.

Chapter 7: Conclusion summarises the contributions, reflects on the broader significance of distribution-free uncertainty quantification for financial machine learning, and outlines directions for future work, including extensions to online conformal prediction, adaptive spatial binning, and integration with richer option pricing models.

Chapter 2

Background and Related Work

This chapter establishes the technical foundations on which the thesis builds. We begin by introducing the financial setting of implied volatility surfaces and the no-arbitrage conditions they must satisfy. We then survey classical and machine learning-based approaches to implied volatility smoothing, situating the Graph Neural Operator method of Gonon, Jacquier, and Wiedemann [10] within this landscape. Finally, we review the theoretical framework of conformal prediction and discuss related work on uncertainty quantification in financial machine learning.

2.1 Implied Volatility and the Options Market

2.1.1 European Options and the Black-Scholes Model

The notation below follows Gonon et al. [10]. At the observation time T_0, a European option has a remaining maturity $\tau = T - T_0$, with expiry $T > T_0$ and strike $K > 0$. Let $F_{T_0,T}$ be the forward price at T_0 for delivery at T. Rather than using the strike directly, we express it relative to this forward price through the log-moneyness $k = \log(K/F_{T_0,T})$. The pair (τ, k) then specifies the option's location in the domain used for volatility modelling.

For a chosen volatility $v > 0$, the Black–Scholes model [5] maps these coordinates to a call price. Dividing the undiscounted price by $F_{T_0,T}$ gives the forward-normalised expression

$$\mathrm{BS}(\tau, k, v) = \Phi(d_1(\tau, k, v)) - e^k \Phi(d_2(\tau, k, v)), \tag{2.1}$$

where Φ denotes the cumulative distribution function of the standard normal distribution and

$$d_1(\tau, k, v) = \frac{-k}{v\sqrt{\tau}} + \frac{1}{2} v\sqrt{\tau}, \qquad d_2(\tau, k, v) = \frac{-k}{v\sqrt{\tau}} - \frac{1}{2} v\sqrt{\tau}. \tag{2.2}$$

Vega measures how the option price changes as volatility changes. For the forward-normalised price above, this sensitivity is

$$\mathcal{V}(\tau, k, v) = \varphi(d_1(\tau, k, v))\sqrt{\tau}, \tag{2.3}$$

where φ denotes the standard normal density [5].

2.1.2 Implied Volatility

The pricing formula can also be used in reverse. Given an observed call price and the remaining model inputs, implied volatility is the value of v that makes the Black–Scholes price match that observation [10]. Writing the undiscounted market price as $C(T, K)$, this means solving

$$C(T, K) = F_{T_0,T} \cdot \mathrm{BS}(\tau, k, v(\tau, k)). \tag{2.4}$$

This inversion places option quotes on a common volatility scale for comparison. Black–Scholes can serve this quoting role even when its assumptions do not fully describe market dynamics [10].

2.1.3 The Implied Volatility Surface

At any given instant, an options market offers contracts across a range of strikes and maturities. Inverting the Black–Scholes formula [5] for each quoted option price yields a corresponding implied volatility. The collection of these values, viewed as a function of time-to-expiry τ and log-moneyness k [9], defines the implied volatility surface. This surface provides a three-dimensional snapshot of how the market prices uncertainty across strikes and maturities.

The implied volatility surface exhibits two well-documented structural features [9]. Across strikes at a fixed maturity, implied volatility typically forms a skewed smile. Across maturities, it displays a term structure that rises or falls with prevailing market conditions. These features summarise how observed option prices vary across the strike-expiry domain and make the surface an important input for derivative pricing, hedging, and portfolio risk measurement.

Because options trade only at a discrete and irregularly spaced set of strikes and maturities, and because this set changes as contracts expire and new ones are listed, the implied volatility surface is not directly observable as a continuous object. It must instead be reconstructed from noisy market quotes through interpolation, extrapolation, and smoothing. Constructing and quantifying the uncertainty of this reconstruction is the central problem this thesis addresses.

2.1.4 No-Arbitrage Conditions

Not every smooth surface defined over the strike-expiry domain corresponds to a financially consistent option-price surface. Static no-arbitrage requirements are commonly expressed through calendar and butterfly conditions, as formalised in Theorem 2.1 of Gonon et al. [10] and related to the SVI framework of Gatheral and Jacquier [9]:

- **No-calendar-arbitrage:** At fixed log-moneyness, total implied variance $\tau v(\tau, k)^2$ must not decrease as maturity increases. This condition corresponds to the required ordering of option prices across maturities [10].
- **No-butterfly-arbitrage:** At each fixed maturity, the corresponding call-option prices must be convex in strike. In implied-volatility terms, this is expressed through a condition involving the smile and its strike derivatives and ensures that the associated risk-neutral density remains non-negative [10].

These conditions motivate the post-processing routine used in this thesis. On a regular grid, the routine applies a fixed-k calendar condition and convexity of total implied variance across neighbouring z points to the lower and upper band surfaces. The latter is a shape regularisation, not the no-butterfly-arbitrage condition of convex call prices in strike. The analytic conditions and the implemented grid constraints are distinguished in Chapter 3.

2.2 Implied Volatility Smoothing

2.2.1 Parametric Approaches

The task of implied volatility smoothing refers to fitting a smooth, arbitrage-free surface $\hat{v}$ to a finite collection $\mathbf{v} = \{v(\tau_i, k_l)\}_{l=1}^{p}$ of observed implied volatilities [10]. Direct interpolation schemes such as

cubic splines can produce smooth fits, but they do not automatically enforce the calendar and butterfly conditions required across the surface.

The industry-standard approach is the Stochastic Volatility Inspired (SVI) model [9], which parameterises total implied variance at each maturity using a five-parameter functional form that captures the characteristic skew and smile. Its extension to full surfaces through the Surface SVI (SSVI) parameterisation [9] provides tractable conditions under which the resulting surface is free of static arbitrage. SVI and its variants have been widely adopted by practitioners.

Despite their practical success, parametric models require a new numerical optimisation whenever the set of market quotes is updated. The resulting calibration depends on the chosen parameterisation, objective function, and optimisation procedure [10].

2.2.2 Neural Network-Based Approaches

The limitations of parametric calibration have motivated a growing body of research on neural network-based implied volatility smoothing. Ackerer et al. [1] introduced deep neural network architectures combined with no-arbitrage soft constraints for volatility surface construction. Their framework also reports uncertainty estimates in regions with few or no observations, although it does not provide conformal coverage guarantees.

A related line of work applies variational autoencoders to volatility smoothing [4], exploiting the fixed-grid structure of FX options markets where strikes are quoted at standard deltas. As noted by Gonon et al. [10], this specificity limits direct application to equity index options such as SPX, where the set of available strikes and maturities changes continuously as contracts expire and new ones are listed.

A further limitation of classical neural network architectures in this setting is the requirement for fixed-size inputs [10]. Options data are inherently irregular: at each point in time, a different collection of strikes and maturities is available. Applying standard feedforward or convolutional networks therefore requires additional preprocessing and a fixed representation of the changing market data.

2.2.3 Operator Deep Smoothing

Gonon, Jacquier, and Wiedemann [10] address the irregular structure of options data by recasting implied volatility smoothing as an operator learning problem. Rather than fitting a parameterised function to each new set of market quotes, they train a map

$$F : \mathcal{A} \to \mathcal{U}$$

between function spaces, where the input $\mathcal{A}$ consists of collections of observed implied volatilities and the output $\mathcal{U}$ is the space of smooth surface functions. At inference time, this operator is evaluated in a single forward pass, bypassing the per-instance optimisation loop that parametric calibration requires.

The market observations are treated as point samples of a continuous volatility function $v : D \to \mathbb{R}$, where D is defined in the transformed coordinates

$$\rho = \sqrt{\tau}, \qquad z = k/\rho,$$

giving $D = (0.01, 1) \times (-1.5, 0.5)$. These coordinates spread the data more uniformly across the input domain [10].

The architecture they adopt is the Graph Neural Operator (GNO) of Li et al. [13]. Its graph-based kernel aggregation can process observations on irregular and changing sets of input locations. Gonon et al. [10] make one structural modification by removing the pointwise linear transformation from the first GNO layer. This allows the first hidden state to be evaluated at output locations that need not

coincide with the observed input locations. Together, the graph construction and this modification enable the model to map changing collections of quoted strikes and maturities to a smoothed surface without fixed-grid preprocessing.

The model is trained on nine years of intraday S&P 500 options data (2012-2020). On the held-out 2021 test year, it achieves substantially lower mean absolute percentage error than the SVI benchmark. The reported calendar and butterfly penalty terms are zero or effectively zero on average across the test set [10].

The trained model is deterministic and returns a single smoothed surface without accompanying prediction intervals. The present thesis complements this surface estimate with conformal uncertainty bands.

2.3 Conformal Prediction

2.3.1 Foundations

Conformal prediction was originally developed by Vovk, Gammerman, and Shafer; the framework is surveyed by Shafer and Vovk [19] and, more recently, by Angelopoulos and Bates [2]. Its central property is the ability to produce prediction sets with finite-sample marginal coverage guarantees without imposing any parametric assumptions on the data-generating process. The key condition required is *exchangeability*: the joint distribution of the calibration and test observations must remain unchanged when their indices are permuted [2]. This condition is natural for observations drawn from a common stable process, while temporally ordered financial data require additional care.

Following the split conformal construction described in Angelopoulos and Bates [2], the procedure works as follows. Given a pre-trained deterministic predictor and a held-out calibration dataset $\{(X_i, Y_i)\}_{i=1}^{n}$, a nonconformity score $A(X_i, Y_i)$ is computed for each calibration point, quantifying how poorly the predictor accounts for that observation. The $(1-\alpha)$-quantile of these n scores, corrected for the finite sample size, is

$$\hat{q}_{1-\alpha} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(\{A(X_i, Y_i)\}_{i=1}^{n}) \,. \tag{2.5}$$

For a new test input X_{n+1}, the prediction set is then

$$\mathcal{C}(X_{n+1}) = \{y : A(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}. \tag{2.6}$$

Under exchangeability of the $n+1$ data points, this construction satisfies the marginal coverage guarantee

$$\mathbb{P}\left(Y_{n+1} \in \mathcal{C}(X_{n+1})\right) \geq 1 - \alpha. \tag{2.7}$$

The guarantee is non-asymptotic: it holds for every finite n and for any choice of underlying predictor and nonconformity score, without requiring the model to be well-specified or calibrated in any distributional sense [19, 2]. Importantly, this is a marginal statement about an individual test observation. In the present application, the conformal intervals are evaluated over the strike-expiry domain and together form complete lower and upper band surfaces around the GNO estimate. The marginal guarantee applies to an individual evaluation location and does not require every location on a daily surface to be covered simultaneously.

2.3.2 Inductive Conformal Prediction

The computationally practical version of conformal prediction is the inductive (or split) conformal predictor [16, 12, 2]. It uses a held-out calibration set that is separate from the training data, computing nonconformity scores only once rather than retraining the predictor for each test point. This separation

makes inductive conformal prediction directly applicable as a post-hoc wrapper around any pre-trained model, which is the approach adopted in this thesis.

2.3.3 Nonconformity Scores

The choice of nonconformity score determines the shape and adaptivity of the resulting prediction sets [2]. The simplest choice for regression is the absolute residual $A(x, y) = |\hat{f}(x) - y|$, which yields symmetric prediction intervals of constant width. More adaptive scores normalise the residual by an estimate of local uncertainty, producing intervals that are wider in regions where the model is less reliable. Conformalized Quantile Regression (CQR) [17] uses quantile regression estimates to produce locally adaptive prediction intervals with valid marginal coverage. The design of economically motivated nonconformity scores tailored to implied volatility is one of the central contributions of this thesis.

2.3.4 Mondrian Conformal Prediction

Mondrian conformal prediction [19, 2] extends the basic framework towards group-conditional coverage by partitioning the input space into disjoint bins and computing a separate conformal quantile within each bin. Under exchangeability, the unmodified Mondrian construction provides marginal coverage conditional on membership in a predefined bin. This is particularly relevant for implied volatility surfaces, where uncertainty varies systematically with liquidity across the strike-expiry domain. In this thesis, spatial binning is used primarily to create adaptive band widths across the surface, and its coverage behaviour is assessed empirically.

2.3.5 Conformal Prediction for Time Series

Applying conformal prediction to financial data requires particular care because financial time series exhibit serial dependence and changing market regimes. As a result, the relevance of older calibration observations can decrease over time. One practical response is to restrict calibration to a sliding window of the most recent observations, so that the conformal quantile reflects current rather than distant historical conditions [3, 2]. Although a rolling window does not by itself establish exchangeability, it makes the calibration responsive to evolving market conditions. Rolling window calibration is therefore used in this thesis, and its coverage in the time-dependent setting is assessed empirically.

2.4 Uncertainty Quantification in Financial Machine Learning

Predictive uncertainty is a broad concern in machine learning and is directly relevant to financial forecasting and risk management. General approaches include Monte Carlo dropout [8], which estimates uncertainty through repeated stochastic forward passes, and deep ensembles [11], which combine predictions from multiple independently trained networks. These methods provide useful uncertainty estimates but require either repeated model evaluations or the training of several models. Conformal prediction offers a model-agnostic alternative that calibrates prediction sets using held-out observations and, under exchangeability, provides finite-sample marginal coverage.

Research connected to options has considered neural networks in several related settings. Wiese et al. [21] use generative models to simulate equity option-market time series, while Ruf and Wang [18] study neural networks for option hedging. These applications differ from constructing uncertainty bands around a smoothed implied volatility surface. More directly, Ackerer et al. [1] construct arbitrage-aware neural-network implied volatility surfaces and quantify prediction uncertainty in sparsely observed regions, but do not use conformal calibration or neural operators. Canete [6] applies online conformal

prediction by treating implied volatility as a point forecast of future realised volatility, rather than constructing bands around a smoothed implied volatility surface. Ma et al. [14] combine conformal prediction with operator learning for scientific and engineering problems, including Darcy flow and car-surface pressure prediction, but not for financial surfaces. Within the literature reviewed in this thesis, no previous study was identified that combines conformal prediction with neural-operator implied volatility smoothing. The present thesis investigates this specific combination.

2.5 Summary

This chapter has established the financial and statistical foundations of the thesis. The implied volatility surface is a central object in options markets and is shaped by calendar and butterfly conditions. Classical parametric methods calibrate this surface instance-by-instance, whereas the operator deep smoothing approach of Gonon et al. [10] learns a mapping from irregular market observations to a smooth surface that can be evaluated in a single forward pass. The resulting GNO estimate is deterministic, and this thesis complements it with conformal uncertainty bands. Under exchangeability, conformal prediction provides a distribution-free framework for constructing prediction sets with finite-sample marginal coverage. Evaluating the calibrated intervals over the strike-expiry domain produces complete lower and upper surfaces around the GNO estimate. Rolling window calibration makes the procedure responsive to changing market conditions, while Mondrian binning provides spatial adaptation across the surface. Their performance in the time-dependent financial setting is evaluated empirically. The following chapter presents the methodology that combines these elements into an uncertainty quantification framework for implied volatility surfaces.

Chapter 3

Methodology

This chapter presents the conformal uncertainty quantification framework developed in this thesis. Section 3.1 introduces the problem setup and notation. Section 3.2 defines the two nonconformity scores used to measure prediction error. Section 3.3 describes the rolling window calibration mechanism and Mondrian spatial binning that adapt the conformal quantile to temporal and spatial heterogeneity. Section 3.4 details the construction of prediction bands in both price and implied volatility space. Section 3.5 presents the finite-grid shape projection applied to the resulting bands and distinguishes its enforced constraints from the no-arbitrage conditions introduced in Chapter 2. Section 3.6 defines the evaluation metrics used to assess the framework in Chapter 5.

3.1 Problem Setup and Notation

Following the transformed coordinate system introduced by Gonon et al. [10] and adopted throughout this thesis, an option quote is identified by a point $x = (\rho, z) \in D$, where $D = (0.01, 1) \times (-1.5, 0.5)$, $\rho = \sqrt{\tau}$ is the square root of time-to-expiry, and $z = k/\rho$ is the maturity-normalised log-moneyness. For a finite and irregular collection of observed quotes, the trained Graph Neural Operator $\hat{F}^{\theta}$ of Gonon et al. [10] produces a smooth implied volatility field $\hat{v} : D \to (0, \infty)$. The GNO is treated in this thesis as a fixed, pre-trained surface estimator. The conformal framework is added as a post-hoc uncertainty layer, without modifying the GNO architecture or its training procedure.

Two market quantities are used to define the nonconformity scores. Let $\mathrm{BS}_{\pm}(x, v)$ denote the forward-normalised, sign-aware Black–Scholes price used by Gonon et al. [10]. The implementation converts this quantity to the corresponding option-price units through

$$P_{t,\pm}(x, v) = D_{t,T} F_{t,T} \, \mathrm{BS}_{\pm}(x, v), \tag{3.1}$$

where $D_{t,T}$ and $F_{t,T}$ are the discount factor and forward price for the relevant maturity. The spread $s_t(x)$ is the observed ask price minus the bid price in the same units as $P_{t,\pm}$. Score B uses the forward-normalised Vega $\mathcal{V}(x, v) = \varphi(d_1(x, v))\rho$. These quantities are built from the Black–Scholes terms introduced in Section 2.1.1, evaluated at $\tau = \rho^2$ and $k = \rho z$.

The framework operates on a temporal sequence of market snapshots. At each time t, a finite set of quote locations $\pi_t \subset D$ is observed together with the corresponding market implied volatilities $v_t(x)$ for $x \in \pi_t$. For the experiment, this set is divided into disjoint context and target sets,

$$\pi_t = \mathcal{C}_t \dot{\cup} \mathcal{T}_t, \qquad |\mathcal{C}_t| \approx 0.8|\pi_t|, \qquad |\mathcal{T}_t| \approx 0.2|\pi_t|. \tag{3.2}$$

The split is made separately within each cell of a 4×4 partition of the (ρ, z) domain. It depends only on the quote coordinates and uses base seed 2026 together with the surface index, making the split

fixed and reproducible for each surface. Approximately 20% of the points in every populated cell are assigned to the target set; a cell containing only one point is kept in the context set.

Only the locations and implied volatilities in $\mathcal{C}_t$ are supplied as observations to the GNO. The coordinates in $\mathcal{T}_t$ are included as query locations, but their implied volatilities are withheld from the model. After the GNO has produced predictions at these locations, the withheld values are used to calculate nonconformity scores and evaluation metrics. The same split is used for Score A and Score B. The target-level bands are subsequently represented on a regular grid when the complete lower and upper surfaces are projected. For a new snapshot at time $t^\star$, the objective is to construct surfaces $v_{\text{lo}} : D \to (0, \infty)$ and $v_{\text{hi}} : D \to (0, \infty)$ around the GNO estimate, with target marginal coverage $1 - \alpha$ and the calendar and total-variance convexity adjustments described in Section 3.5.

3.2 Nonconformity Scores

The choice of nonconformity score determines how a raw prediction error is translated into a measure of nonconformity and consequently how the resulting bands are shaped across the domain. Two scores are used. Score A normalises a price residual by the prevailing bid-ask spread, while Score B weights an implied volatility residual by the option's Vega. In both cases, the rescaling varies across the surface and reflects differences in local liquidity or price sensitivity.

3.2.1 Score A: Spread-Normalised Price Error

The first score normalises the price residual by the prevailing bid-ask spread, so that an error is judged relative to the width of the observed market quote:

$$A_t^{\text{spr}}(x) = \frac{2\,|r_t^P(x)|}{\max\{s_t(x), 0.10\}}, \qquad r_t^P(x) = P_{t,\pm}(x, \hat{v}_t(x)) - P_{t,\pm}(x, v_t(x)), \qquad x \in \mathcal{T}_t. \tag{3.3}$$

This construction adapts the spread-relative pricing error used by Gonon et al. [10] for conformal calibration. Here, $P_{t,\pm}$ is the option price defined in Equation 3.1, and $s_t(x)$ is measured in the same option-price units. The floor of 0.10 avoids unstable normalisation when the recorded spread is very small or zero. The market reference $P_{t,\pm}(x, v_t(x))$ is reconstructed from the stored market IV using the same forward and discount factor as the predicted price. The stored IV is obtained from the quoted midpoint using the forward and a zero-rate inversion, so this reconstructed reference need not equal the quoted midpoint. Score A is therefore interpreted as a spread-scaled difference between the two reconstructed prices, rather than as a direct test of whether the predicted price lies within the recorded bid–ask interval.

3.2.2 Score B: Vega-Weighted Implied Volatility Error

The second score operates directly in implied volatility space and multiplies the residual by a weight derived from the option's forward-normalised Vega. Options with above-average Vega therefore receive greater weight:

$$A_t^{\text{IV}}(x) = w_V(x; \hat{v}_t)\,|r_t^{\text{IV}}(x)|, \qquad r_t^{\text{IV}}(x) = \hat{v}_t(x) - v_t(x), \qquad x \in \mathcal{T}_t, \tag{3.4}$$

where the implemented weight is

$$w_V(x; \hat{v}_t) = \max\left\{ \frac{\mathcal{V}(x, \hat{v}_t(x))}{\frac{1}{|\mathcal{T}_t|}\sum_{x' \in \mathcal{T}_t} \mathcal{V}(x', \hat{v}_t(x'))}, 1 \right\}. \tag{3.5}$$

For both calibration and prediction, Vega is evaluated at the GNO estimate and rescaled by the mean prediction-based Vega over the target set of the surface. Options above that mean receive a weight greater than 1, while the floor keeps the remaining residuals on their original scale. The observed market implied volatility enters the calibration residual only. This prediction-based weighting replaces the market-IV-based calibration weight proposed in the research exposé: it makes the weight available before the target IV is observed and lets the Vega-IV band invert the same score that was calibrated.

3.3 Rolling Window Calibration and Mondrian Binning

Financial markets are non-stationary: the distribution of prediction errors observed in one period need not remain unchanged in another. Let L denote the maximum number of surfaces in the rolling window. The framework restricts calibration to the L most recently processed market surfaces rather than using the complete history. Let the available snapshots be ordered by time as $t_1 < t_2 < \cdots$. For an evaluation at t_j, the window W_j contains up to the L most recent available snapshots preceding t_j. The window counts observed surfaces rather than elapsed calendar days, so gaps between data-collection dates do not alter its definition.

The target scores from the surfaces in W_j are pooled into

$$\mathcal{A}(W_j) = \{A_t(x) : t \in W_j,\ x \in \mathcal{T}_t\}. \tag{3.6}$$

Let $n = |\mathcal{A}(W_j)|$ and, when $n \geq 1$, let $A_{(1)} \leq \cdots \leq A_{(n)}$ denote the ordered pooled scores. The implementation uses the finite-sample conformal threshold

$$\hat{q}_{1-\alpha} = \begin{cases} A_{(r_n)}, & r_n \leq n, \\ +\infty, & r_n > n, \end{cases} \qquad r_n = \lceil (n+1)(1-\alpha) \rceil . \tag{3.7}$$

The infinite threshold applies when the available scores do not include the required order statistic. Under exchangeability, this is the standard finite-sample quantile used to obtain marginal coverage at level $1-\alpha$ [19, 2]. In the walk-forward evaluation, the target points on each new surface are evaluated using only the preceding window. Their scores are added to the window afterwards, allowing the surface to contribute to subsequent predictions without calibrating its own bands.

A single quantile computed over the entire domain D does not reflect possible variation in the score distribution across maturities and moneyness levels. To introduce spatial adaptation, the framework applies *Mondrian conformal prediction* [20, 19]: the domain is partitioned into a regular grid of $B_\rho \times B_z$ bins, and Equation 3.7 is applied separately to the scores in each sufficiently populated bin. An evaluation point receives the quantile of the bin containing its (ρ, z) coordinates.

Because each bin contains fewer scores than the pooled calibration set, two stabilising rules are used. A bin receives its own quantile only when it contains at least 10 calibration scores; otherwise, it uses the global quantile from the same window. For a sufficiently populated bin, the raw bin quantile $q_{\text{bin}}^{\text{raw}}$ is combined with the global quantile using

$$q_{\text{bin}} = \lambda\, \hat{q}_{1-\alpha} + (1-\lambda)\, q_{\text{bin}}^{\text{raw}}. \tag{3.8}$$

The experiments use $L = 5$ surfaces, a 4×4 partition, and $\lambda = 0.3$. This shrinkage is an empirical regularisation that reduces unstable differences between bin estimates and the global value; the coverage of the resulting spatially adaptive bands is therefore evaluated empirically. Points outside the bounds of D are assigned the global quantile.

3.4 Prediction Bands

At a new prediction time $t^\star$, the context observations are supplied to the GNO and it evaluates $\hat{v}_{t^\star}(x)$ at the held-out target coordinates $x \in \mathcal{T}_{t^\star}$. Each target location is assigned the corresponding global or Mondrian conformal quantile. Two modes of constructing prediction bands around $\hat{v}_{t^\star}(x)$ are considered, corresponding to the two nonconformity scores of Section 3.2. The resulting target-level bounds are subsequently represented on a regular grid for the surface projection described in Section 3.5.

3.4.1 Price-Spread Mode

Under Score A, the conformal quantile is used to construct a half-width in price space, scaled by the spread at each quote:

$$h^P(x) = \tfrac{1}{2}\, q^{\mathsf{spr}}_{1-\alpha}(x)\, \max\{s_{t^\star}(x),\, 0.10\}\,, \tag{3.9}$$

giving price bands

$$P_{\mathsf{lo}}(x) = P_{t^\star,\pm}(x, \hat{v}_{t^\star}(x)) - h^P(x), \qquad P_{\mathsf{hi}}(x) = P_{t^\star,\pm}(x, \hat{v}_{t^\star}(x)) + h^P(x). \tag{3.10}$$

Here, $P_{t^\star,\pm}$ is the option-price function defined in Equation (3.1), using the forward and discount factor for the quote. Corresponding bands in implied volatility space are obtained by inverting the same pricing relation at the two endpoints.

3.4.2 Vega-IV Mode

Under Score B, bands are constructed directly in implied volatility space. The conformal quantile is converted to an implied volatility half-width via the same prediction-based Vega weighting used in the nonconformity score:

$$h_v(x) = \frac{q^{\mathsf{IV}}_{1-\alpha}(x)}{w^\star_V(x)}, \qquad w^\star_V(x) = w_V(x; \hat{v}_{t^\star}), \tag{3.11}$$

giving implied volatility bands

$$v_{\mathsf{lo}}(x) = \max\Big\{\hat{v}_{t^\star}(x) - h_v(x),\, 10^{-4}\Big\}\,, \qquad v_{\mathsf{hi}}(x) = \hat{v}_{t^\star}(x) + h_v(x), \tag{3.12}$$

where 10^{-4} is the numerical floor used in the implementation to keep the lower endpoint strictly positive. If required, the corresponding price endpoints are recovered as $P_{\mathsf{lo}}(x) = P_{t^\star,\pm}(x, v_{\mathsf{lo}}(x))$ and $P_{\mathsf{hi}}(x) = P_{t^\star,\pm}(x, v_{\mathsf{hi}}(x))$. Both band modes produce implied-volatility lower and upper surfaces that can be passed through the same projection described next. Their empirical comparison in Chapter 5 motivates the use of Score B as the primary construction in this thesis.

3.5 Finite-Grid Shape Projection

The prediction bands constructed in Section 3.4 are designed to provide statistical coverage around the GNO surface. A post-processing step imposes fixed-k calendar monotonicity and total-variance convexity on a finite grid for the lower and upper bands. This section first summarises the no-arbitrage conditions from the base GNO paper [10] and then describes the narrower conditions enforced by the implemented shape projection.

3.5.1 Calendar and Butterfly Constraints

Theorem 3.1 (Adapted from Theorem 2.1 of Gonon et al. [10]). *Let $\hat{v} : (0,\infty) \times \mathbb{R} \to (0,\infty)$ be continuous and satisfy:*

(i) ***Calendar arbitrage.*** *For each log-moneyness $k \in \mathbb{R}$, the map $\tau \mapsto \hat{v}(\tau,k)\sqrt{\tau}$ is non-decreasing and vanishes at $\tau = 0$.*

(ii) ***Butterfly (strike) arbitrage.*** *For every $\tau > 0$, the maturity slice $\hat{v}_\tau := \hat{v}(\tau,\cdot)$ is of class C^2 and satisfies*

$$\mathit{But}(\tau,\cdot,\hat{v}_\tau,\,\partial_k\hat{v}_\tau,\,\partial_k^2\hat{v}_\tau) \geq 0, \tag{3.13}$$

where, for $v_0 = \hat{v}_\tau(k)$, $v_1 = \partial_k\hat{v}_\tau(k)$, $v_2 = \partial_k^2\hat{v}_\tau(k)$,

$$\mathit{But}(\tau,k,v_0,v_1,v_2) = \Big(1 + d_1(\tau,k,v_0)\,v_1\sqrt{\tau}\Big)\Big(1 + d_2(\tau,k,v_0)\,v_1\sqrt{\tau}\Big) + v_0\,v_2\,\tau. \tag{3.14}$$

It also satisfies the asymptotic condition

$$\limsup_{k\uparrow\infty} \frac{\hat{v}_\tau(k)^2}{k} < \frac{2}{\tau}. \tag{3.15}$$

Then $(T,K) \mapsto BS(\tau,k,\hat{v}(\tau,k))$ defines a call price surface free of static arbitrage.

Condition (i) is the calendar spread condition already introduced in Section 2.1.4: total implied variance should not decrease as maturity increases at the same log-moneyness. Condition (ii) is the analytic butterfly condition, for which a non-negative But functional corresponds to a non-negative risk-neutral density. The asymptotic condition concerns the behaviour of the surface outside the bounded domain used in this thesis and is therefore not part of the finite-grid post-processing step.

The base paper incorporates the calendar and butterfly conditions during GNO training through the soft penalties $\mathcal{L}_{\text{cal}}$ and $\mathcal{L}_{\text{but}}$, numbered Equations (11) and (10), respectively, in that paper. The pre-trained GNO is not retrained in this thesis. After band construction, the conformal lower and upper surfaces are adjusted using a fixed-k calendar condition and convexity of total implied variance across z. The latter is a linear shape regularisation; it does not enforce the paper's butterfly functional or convexity of call prices in strike.

3.5.2 Projection Algorithm

The lower and upper band surfaces are adjusted jointly. The calculations are performed in terms of *total implied variance*

$$w(\rho,z) := v(\rho,z)^2\rho^2, \qquad \tau = \rho^2, \quad k = \rho z. \tag{3.16}$$

For consecutive maturity coordinates $\rho_i < \rho_{i+1}$ and a grid point z_j on the longer-maturity row, the point with the same log-moneyness k on the shorter-maturity row is

$$z_i^{(j)} = \frac{\rho_{i+1}}{\rho_i} z_j. \tag{3.17}$$

When $z_i^{(j)}$ lies inside the grid, the calendar condition requires

$$w\Big(\rho_i, z_i^{(j)}\Big) \leq w(\rho_{i+1}, z_j). \tag{3.18}$$

The shorter-maturity value is obtained by linear interpolation in z. This is the fixed-k comparison used in the GNO paper because $\rho_i z_i^{(j)} = \rho_{i+1} z_j$. The implementation interpolates total variance, which gives a conservative discrete form of the corresponding positive total-volatility condition.

The total-variance convexity condition compares three neighbouring z points at each fixed maturity. The total variance at the middle point must not exceed the linearly interpolated value between the two outer points. For $z_1 < z_2 < z_3$,

$$w(\rho, z_2) \leq \frac{\Delta z_2\, w(\rho, z_1) + \Delta z_1\, w(\rho, z_3)}{\Delta z_1 + \Delta z_2}, \qquad \Delta z_1 = z_2 - z_1, \quad \Delta z_2 = z_3 - z_2. \tag{3.19}$$

At fixed ρ, $k = \rho z$ is linear, but strike $K = F\exp(k)$ is not linear in z. Convexity of total variance across z therefore does not imply convexity of call prices across strikes. Equations 3.18 and 3.19 are the two finite-grid conditions enforced by the projection; satisfying them does not establish butterfly no-arbitrage.

The held-out target quotes lie at irregular (ρ, z) locations. For each evaluation surface, their band values are mapped by linear interpolation to a rectangular grid formed from the distinct target ρ values, rounded to four decimal places, and 50 equally spaced z values. Nearest-neighbour interpolation fills grid locations outside the convex hull. The adjusted grid values are obtained from one sparse linear program that minimises the total absolute change in both surfaces:

$$\min_{\widetilde{w}_{\mathrm{lo}}, \widetilde{w}_{\mathrm{hi}}} \sum_{b \in \{\mathrm{lo},\mathrm{hi}\}} \sum_{i,j} \left| \widetilde{w}_b(\rho_i, z_j) - w_b(\rho_i, z_j) \right|. \tag{3.20}$$

The optimisation is subject to Equations 3.18 and 3.19 for both surfaces, positivity, and the ordering constraint $\widetilde{w}_{\mathrm{lo}} \leq \widetilde{w}_{\mathrm{hi}}$ at every grid point. Absolute changes are represented with auxiliary variables, so the problem remains linear and is solved using the HiGHS solver. The corrected total variances are then converted back to implied volatility using $\widetilde{v} = \sqrt{\widetilde{w}/\rho^2}$. This is summarised in Algorithm 1.

Algorithm 1 Conformal bands with finite-grid shape projection

Require: Trained GNO $\hat{F}^{\theta}$, rolling window W, spatial binning of D, tolerance ε, coverage level $1-\alpha$

1: split each surface into context set $\mathcal{C}_t$ and target set $\mathcal{T}_t$ using Equation 3.2
2: supply the context locations and IV values to the GNO; include the target coordinates as query locations without their IV values
3: **Calibration:** compute target-set nonconformity scores (Equation 3.3 and/or Equation 3.4) in each bin; store the empirical quantile $\hat{q}_{1-\alpha}$ per Equation 3.7
4: **Predict:** evaluate $\hat{v} = \hat{F}^{\theta}$ at the target locations; compute $v_{\mathsf{lo}}, v_{\mathsf{hi}}$ via the price-spread or Vega-IV mode
5: **Interpolate:** map both IV bands to the rectangular (ρ, z) grid
6: construct the fixed-k calendar constraints using Equation 3.17
7: construct the total-variance convexity constraints from neighbouring z points at each maturity
8: jointly minimise Equation 3.20 subject to both shape conditions, positivity, and lower/upper ordering
9: **Map back:** interpolate both projected surfaces to the target locations
10: **Check:** count remaining fixed-k calendar and total-variance convexity violations on the grid and lower/upper crossings at the target locations
11: **return** projected IV bands $(\tilde{v}_{\mathsf{lo}}, \tilde{v}_{\mathsf{hi}})$ and projection diagnostics

The projected grid values are mapped back to the target locations by linear interpolation; if a mapped value is unavailable, the original band endpoint at that target is retained. The joint optimisation imposes lower/upper ordering directly on the grid, and crossings are also checked after mapping back. Corresponding price endpoints can, when required, be recovered using the option-price function $P_{t^\star,\pm}$ defined in Equation (3.1).

3.5.3 Coverage after Projection

The direct conformal construction provides the bands before projection. Since the grid mapping and the optimisation can each adjust their endpoints, coverage after projection is evaluated empirically rather than inferred from the pre-projection statement. To separate these effects, the evaluation also maps the unadjusted grid bands back to the target quotes using the same linear interpolation and original-endpoint fallback as for the projected bands. The resulting interpolation-only bands omit the optimisation. Direct, interpolation-only, and projected bands are compared at identical target quotes for empirical coverage, mean IV width, endpoint movements, and containment of the GNO point prediction. The evaluation also reports the remaining fixed-k calendar and total-variance convexity counts and the number of lower–upper crossings.

3.6 Evaluation Metrics

The framework is evaluated in walk-forward order. Each evaluation surface is assessed using only the preceding surfaces in the rolling calibration window and is added to the window afterwards. Consequently, an evaluation surface is never used to calibrate its own prediction bands.

Coverage: For an evaluation surface at $t^\star$, within-surface coverage is the fraction of held-out target implied volatilities contained in the band:

$$\widehat{\mathrm{Cov}}_{t^\star} = \frac{1}{|\mathcal{T}_{t^\star}|} \sum_{x \in \mathcal{T}_{t^\star}} \mathbf{1}\{v_{t^\star}(x) \in [v_{\mathsf{lo}}(x), v_{\mathsf{hi}}(x)]\}\,. \tag{3.21}$$

It is compared with the nominal target $1-\alpha$. Coverage is computed separately for the global and Mondrian quantiles and, where relevant, before and after projection. The per-surface values are then averaged across the evaluation period with equal weight given to each surface, rather than pooling all evaluation quotes into one summary.

Sharpness: The IV band width is $v_{\mathsf{hi}}(x) - v_{\mathsf{lo}}(x)$. Its mean and median are calculated over the target locations of each evaluation surface after the projected grid has been mapped back to those locations. Narrower bands are preferred when coverage is comparable.

Calibration diagnostics: The global quantile and the mean, minimum, and maximum Mondrian quantiles are recorded as the rolling window advances. The number of calibration scores and the mean score per surface are also retained. These quantities show how the calibration threshold changes through time and across regions of the (ρ, z) domain.

Calendar and total-variance convexity diagnostics: The fixed-k calendar and total-variance convexity violations defined by the grid-based checks of Section 3.5.2 are counted before and after projection for the lower and upper band surfaces. The evaluation also records whether the projection converged and counts lower–upper crossings on the grid and at the target locations.

Point-prediction error: The mean absolute error between the GNO prediction $\hat{v}_{t^\star}(x)$ and the held-out market implied volatility $v_{t^\star}(x)$, for $x \in \mathcal{T}_{t^\star}$, is recorded as a common reference value. Because both score modes use the same trained GNO predictions and target masks, differences in their coverage and sharpness arise from the nonconformity score and band construction rather than from different point predictions.

3.7 Summary

This chapter has presented the methodology for conformal uncertainty quantification around GNO-based implied volatility surfaces. Each market surface is divided into context observations for the

GNO and held-out target locations for conformal calibration and evaluation. The framework combines two economically motivated nonconformity scores, walk-forward rolling calibration, Mondrian spatial binning, prediction bands in price and implied-volatility space, and the calendar and total-variance convexity projection of the lower and upper band surfaces. Its evaluation uses matched coverage, sharpness, calibration, projection, crossing, and point-prediction diagnostics. These components provide the methodological basis for assessing the six research objectives set out in Chapter 1. The following chapter describes their software implementation, and Chapter 5 reports the empirical comparison of Score A and Score B on the S&P 500 options data.

Chapter 4

Implementation

4.1 Software Architecture

The conformal uncertainty quantification framework is implemented as a Python package, `volatility_smoothing/conformal/`, together with the experiment notebook. It operates as a post-hoc layer around the pre-trained GNO, whose architecture and weights are not modified. The implementation constructs masked GNO inputs, receives predictions at held-out target locations, and returns calibrated prediction bands. The package is structured into six modules, each covering one part of the pipeline described in Chapter 3. Table 4.1 lists the main interfaces and their roles.

Module	Primary interface	Role
`nonconformity_scores.py`	`SpreadNormalizedScore` `VegaNormalizedScore`	Score A: spread-normalised price error Score B: Vega-weighted IV error
`rolling_window.py`	`RollingWindowCalibration`	FIFO window of target scores from the L most recent surfaces
`mondrian.py`	`MondrianConformal`	Spatial binning and per-bin quantile computation
`prediction_bands.py`	`PredictionBands`	Band construction in price-spread and Vega-IV modes
`no_arbitrage_projection.py`	`project_band_surfaces_fixed_k`	Joint fixed-k calendar and total-variance convexity adjustment of both band surfaces
`evaluation.py`	`build_masked_gno_input`	Context/target masking, GNO query construction, and surface-level bootstrap summaries

Table 4.1: Main modules of the `volatility_smoothing/conformal/` package and their roles.

The modules are wired together in `conformal_prediction.ipynb`, which is the integration point for the reported experiment and is described in Section 4.9. The notebook also defines the exact conformal order statistic, the Score B band helper, and the reported projection diagnostics. The shared prediction-based Score B weighting is implemented in `nonconformity_scores.py`. Each component is described below.

4.2 Nonconformity Score Computation

Both nonconformity scores are implemented in `nonconformity_scores.py` as subclasses of a common `NonconformityScore` base class. Its `compute` method accepts the GNO predictions, the correspond-

ing market implied volatilities, and the auxiliary market data required by the score. The dispatcher `compute_scores_from_data` selects the scorer from the `score_type` argument. In the final experiment, it receives only the held-out target subset and the GNO predictions at those target locations.

Score A (`SpreadNormalizedScore`) works in price space. It takes three steps: (1) compute predicted and reference prices from the corresponding implied volatilities using the Black-Scholes formula in transformed coordinates - $d_1 = -z/v + v\rho/2$ and $d_2 = -z/v - v\rho/2$, with call and put prices handled separately based on the sign of the log-moneyness $k = \rho z$, (2) read bid and ask prices directly from the dataloader as dollar-denominated quantities and floor the spread at 0.10, (3) divide twice the absolute price residual by the floored spread, per Equation 3.3. The reference price is reconstructed from the stored market IV, while the recorded bid and ask determine the scale of the score. Accordingly, the score measures a reconstructed price difference relative to the observed spread; it is not a direct bid–ask containment indicator.

Score B (`VegaNormalizedScore`) works directly in implied volatility space. The notebook helper `score_from_data` delegates to `compute_scores_from_data`. Both it and the band constructor use `prediction_vega_weights`, which computes forward-normalised Vega at the GNO-predicted IV and divides by its mean over the held-out target set. The completed ratio $\mathcal{V}/\bar{\mathcal{V}}$ is floored at `min_weight`$=1.0$, so options with above-average prediction-based Vega are up-weighted and no option is weighted below 1. The observed target IV enters only the calibration residual; the weight used to construct a new band is calculated from the GNO estimate in the same way as the calibration weight.

4.3 Rolling Window Calibration

The rolling window is implemented in `rolling_window.py` as a `RollingWindowCalibration` class that maintains an ordered list of `CalibrationSurface` dataclasses. Each entry stores the target (ρ, z) coordinates and target nonconformity scores from one market snapshot. The window operates as a FIFO buffer of fixed size L: `add_surface` appends a new surface and discards the oldest if the list exceeds L entries. `get_calibration_data` concatenates the target coordinates and scores currently in the window for global and Mondrian quantile computation. The window counts available snapshots rather than elapsed time.

Two state queries are available: `is_ready` returns true as soon as at least one surface has been added, and `is_full` returns true once the window has reached capacity L. In the experiment pipeline, quantile computation begins immediately after the first surface is added, so the Mondrian calibration runs on a growing window during the early calibration surfaces and on a full window of L surfaces thereafter.

4.4 Mondrian Spatial Binning

Mondrian binning is implemented in `mondrian.py` as a `MondrianConformal` class. At initialisation, the domain $D = (0.01, 1) \times (-1.5, 0.5)$ is partitioned into a regular $B_\rho \times B_z$ grid of `MondrianBin` objects using evenly spaced edges along each axis. Each bin stores the target nonconformity scores from the rolling window that fall within its boundaries.

Calibration proceeds in two passes. The final experiment uses `ExactOrderStatisticMondrian`, a subclass defined in `conformal_prediction.ipynb`, to apply the finite-sample conformal order statistic in Equation 3.7. First, the global quantile is computed over all scores in the window as a fallback. Second, per-bin quantiles are computed for bins that contain at least `min_samples_per_bin` $= 10$ scores; bins below this threshold are assigned the global quantile directly. For bins that meet the threshold, the raw bin quantile is shrunk towards the global quantile by a factor λ:

$$q_{\text{bin}} = \lambda\, \hat{q}_{1-\alpha} + (1 - \lambda)\, q_{\text{bin}}^{\text{raw}}, \tag{4.1}$$

with $\lambda = 0.3$ in the experiments of Chapter 5. This prevents bins that marginally exceed the minimum count from producing extreme quantile estimates while preserving the spatial adaptivity of well-populated bins.

At evaluation time, `get_quantiles` assigns each target point the quantile of the bin containing its (ρ, z) coordinates. Points outside the domain boundaries receive the global quantile. Bin lookup uses `np.searchsorted` on the precomputed bin edges.

4.5 Prediction Band Construction

Band construction is provided by `prediction_bands.py` through the `PredictionBands` class. It accepts the GNO target predictions, the per-target global or Mondrian conformal quantiles, and the market data required by the chosen mode, and returns lower and upper bounds in IV and price space. In the final pipeline, Score A uses `PredictionBands.compute`, while Score B uses the notebook helper `vega_iv_bands`. Both the notebook helper and the package interface use the shared prediction-based Vega weight, so band construction inverts the nonconformity score used for calibration.

Under the Vega-IV mode, the helper takes four steps: (1) compute forward-normalised Vega at each target using the GNO-predicted IV, (2) form the same relative weight $w_V = \mathcal{V}/\bar{\mathcal{V}}$ used during calibration over the target set and floor it at `min_vega`$= 1.0$, (3) divide the per-target conformal quantile by the weight to obtain the IV half-width $h_v = q/w_V$, and (4) subtract and add h_v from the GNO prediction to obtain v_{lo} and v_{hi}, with v_{lo} clipped at 10^{-4}. Price bands are recovered by evaluating the Black-Scholes formula at the two IV endpoints.

Under the price-spread mode, `compute` takes three steps: (1) compute the GNO predicted price using the Black-Scholes formula, (2) read bid and ask prices from the dataloader and floor the spread at `min_spread`$= 0.10$, (3) add and subtract the price half-width $h_P = (q/2) \cdot s$ from the predicted price to get P_{lo} and P_{hi}. IV bands are then obtained by inverting the Black-Scholes price at each endpoint via `implied_volatility_from_price`, which root-finds per point using Brent's method over the interval $[10^{-4}, 5.0]$. Points where no sign change is found within this interval are clipped to the nearest bound rather than raising an error.

One important distinction is that the `black_scholes_price` function in this module produces dollar-denominated prices - that is, discount $\times$ forward $\times$ BS(τ, k, v) - rather than the forward-normalised, dimensionless BS$_{\pm}(x, v)$ used in Chapter 3. These currency units are used consistently with the observed bid and ask prices in Score A. Score B is constructed directly in implied-volatility space using the forward-normalised Vega weights described above.

4.6 Finite-Grid Shape Projection

The finite-grid shape projection is implemented in `no_arbitrage_projection.py`. The final notebook uses `check_butterfly_arbitrage`, `check_calendar_arbitrage_fixed_k`, and `project_band_surfaces_fixed_k`. The scattered target-band values are represented on a rectangular grid whose rows correspond to the available target ρ values, with maturities $\tau = \rho^2$, and whose columns contain 50 evenly spaced z values. The mapping uses linear interpolation from `scipy.interpolate.griddata`, with nearest-neighbour interpolation outside the convex hull.

The two check functions each return an `ArbitrageConstraints` dataclass recording the violation count and maximum violation magnitude. Despite its historical name, `check_butterfly_arbitrage` implements only the total-variance convexity check, not a test of call-price convexity in strike. It iterates over each maturity slice, computes total variance $w = v^2\tau$ across the z direction, and flags a middle grid point when its total variance exceeds the linearly interpolated value between its two

neighbours by more than a tolerance $\varepsilon = 10^{-6}$. `check_calendar_arbitrage_fixed_k` compares consecutive maturity rows at the same log-moneyness $k = \rho z$. For a longer-maturity grid coordinate z_j, the matching shorter-maturity coordinate is $\rho_{i+1} z_j / \rho_i$ and its total variance is obtained by linear interpolation.

`project_band_surfaces_fixed_k` expresses the fixed-k calendar condition and the total-variance convexity condition as sparse linear inequalities in total variance. It then calls `scipy.optimize.linprog` with the HiGHS solver to minimise the sum of absolute changes from the original lower and upper grids. Both surfaces are solved together, with positive total variance and lower/upper ordering included as additional constraints. Before conversion back to implied volatility, values that fall marginally below the small positive variance floor because of numerical solver tolerance are set to that floor, and the number of these adjustments is recorded. The function returns the projected surfaces together with the initial and final violation counts, solver status, iteration count, projection distance, floor-adjustment count, and crossing count.

The corrected grid values are mapped back to the target (ρ, z) coordinates through a second `scipy.interpolate.griddata` call. A target outside the grid convex hull retains its corresponding pre-projection endpoint. The resulting values form the final $\tilde{v}_{\text{lo}}$ and $\tilde{v}_{\text{hi}}$ used for coverage and band-width evaluation in Chapter 5. Lower/upper crossings are counted after mapping back.

4.7 Data Acquisition and Preprocessing

The SPX option chains used in the empirical study were collected through the OpenBB Python interface using its Cboe provider and the symbol `SPX` [15]. A scheduled retrieval script requested the available option chain and saved the returned expiration dates, strikes, option types, bid prices, and ask prices as CSV files. The files were written in the column format expected by the existing `WRDSOptionsDataset` loader. The name of this loader therefore refers to the compatible file schema and does not imply that the observations were obtained from WRDS.

During this transformation, the calendar date on which the retrieval completed was stored in the `date` field. The dates used to order the surfaces in the notebook and figures are consequently collection dates rather than separate exchange-provided quote timestamps. The `am_settlement` field was set to one as a compatibility field required by the loader's filtering step; it is not used in this thesis as an empirical classification of the settlement convention of each individual contract.

The loader parses the stored collection and expiry dates as date-only values, represented at 00:00 without a time zone. It computes time to maturity as $\tau = (\text{expiry date} - \text{collection date})/365$ years, where the numerator is the number of elapsed calendar days. The 00:00 representation is a computational convention for these fields; it does not assert that an option quote was observed or a contract settled at midnight. Because the files contain no actual quote or settlement time, the intraday component of remaining maturity is unavailable.

For each collection date and expiry, the pipeline first forms the arithmetic midpoint of each bid and ask pair. It then estimates the discount factor D and forward F from paired call and put midpoints at the same strikes using the put–call parity regression $C_{\text{mid}}(K) - P_{\text{mid}}(K) = a + bK$: the fitted values are $D = -b$ and $F = a/D$. The default selection takes up to three strikes with $P_{\text{mid}} > C_{\text{mid}}$ and up to three with $P_{\text{mid}} \leq C_{\text{mid}}$, taking the first available strikes in ascending order within each group. These are estimates from the stored option quotes, not independently observed forward or discount curves; without per-contract quote times, the simultaneity of the paired midpoints cannot be verified. The pipeline then computes log-moneyness and implied volatility and restricts the observations to the transformed GNO domain.

The raw directory contains 56 files. Ten files whose stored dates do not correspond to a regular Cboe trading session are excluded from the analysis: eight have weekend dates, while 25 December 2025

and 3 July 2026 are exchange holidays. The 24 December 2025 early-close file is retained. The original files are not deleted; an explicit date list filters the dataset index in memory before the GNO dataset is constructed. This leaves 46 analysis surfaces with collection dates from 11 December 2025 to 11 August 2026.

The audit found no duplicate rows or duplicate contract keys within a file and no completely identical files. The 25 December file repeats the bid and ask values of the contracts also present on 26 December; excluding the non-regular-session file prevents both snapshots from entering the analysis. The retained collection dates are irregular because no successful file is available for every scheduled date. Provider API and integration errors occurred during collection. The CSV files also do not contain the provider quote timestamp, retrieval time, or time zone, so the collection dates serve both to order the available snapshots and to calculate the date-based maturities described above.

4.8 Context and Target Construction

The masking helpers are implemented in `evaluation.py`. For each retained surface, `stratified_context_target_indices` assigns approximately 20% of the quotes to the target set within each cell of the same 4×4 (ρ, z) partition used for Mondrian calibration. The random generator uses seed $2026 + i$ for surface index i. A cell containing only one quote remains in the context set, and assertions verify that the context and target indices are non-empty, disjoint, and together contain every quote.

`build_masked_gno_input` passes only the context features and context IV values through the GNO input arrays `x` and `pos_x`. It appends the target coordinates to `pos_y` as query locations and rebuilds the graph edges, without adding target IV values to the model input. `read_masked_gno_output` then separates the target predictions from the GNO's regular-grid outputs. The target subset is passed to the score, band, coverage, width, and point-error calculations. Score A and Score B use the same indices on every surface.

4.9 Experiment Notebook

The experiment notebook `conformal_prediction.ipynb` is the integration point for the reported pipeline. It loads the pre-trained GNO checkpoint and the WRDS-compatible OpenBB files through `WRDSOptionsDataset` and `GNOOptionsDataset`, applies the ten-date filter, and divides the 46 retained surfaces into 15 initial calibration surfaces and 31 walk-forward evaluation surfaces. The fixed settings used in the final run are listed in Table 4.2.

Hyperparameter	**Symbol**	**Value**
Coverage level	$1 - \alpha$	90%
Rolling window size	L	5 surfaces
Mondrian bins	$B_\rho \times B_z$	4×4
Shrinkage factor	λ	0.3
Min samples per bin		10
Held-out target fraction		20%
Mask base seed		2026
Surface bootstrap resamples		$2{,}000$
Surface bootstrap seed		2026

Table 4.2: Fixed settings used in the final experiment.

The pipeline is encapsulated in `run_conformal_pipeline`, which accepts `'vega'` for Score B or `'spread'` for Score A. Each of the 15 initial calibration surfaces is masked, the GNO predicts its target points from the context observations, and the resulting target scores are added to the rolling window. During walk-forward evaluation, the global and Mondrian quantiles are first obtained from the preceding window. The GNO then predicts the current target points, the two band variants are constructed and projected, and the target coverage, width, point-prediction error, fixed-k calendar and total-variance convexity counts, and lower/upper crossings are recorded. Only after this evaluation are the current target scores added to the rolling window. The function is run once for each score with identical masks.

The main coverage and width summaries first calculate one value for each of the 31 evaluation surfaces and then average those values, giving equal weight to each surface. The reported 95% bootstrap intervals use `surface_bootstrap_mean_ci` with 2,000 resamples and seed 2026. Each resampling unit is a complete evaluation surface rather than an individual quote.

Figure 4.1 shows the 46 retained surfaces. The first 15 surfaces, from 11 December 2025 to 17 February 2026, form the initial calibration set, while the remaining 31 surfaces, from 3 March to 11 August 2026, are evaluated in walk-forward order. Each retained surface contains between 8,677 and 11,325 quotes, with a mean of 10,119. The context sets contain 6,943 to 9,058 points and the held-out target sets contain 1,734 to 2,267 points.

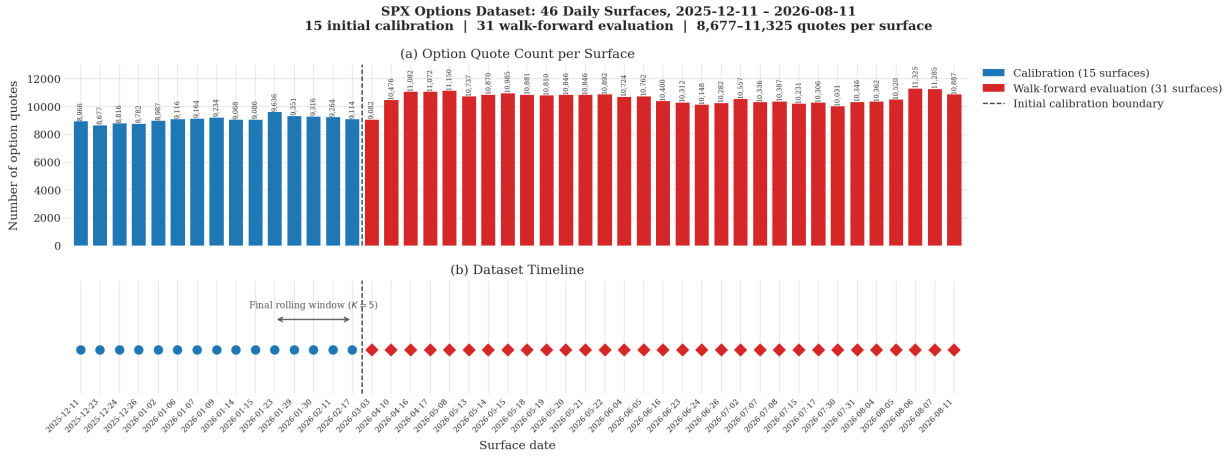


Figure 4.1: The 46 retained SPX option surfaces after the regular-session-date filter. Blue indicates the 15 initial calibration surfaces (indices 0–14), and red indicates the 31 walk-forward evaluation surfaces (indices 15–45).

4.10 Summary

This chapter described the implementation of the conformal framework as a Python package and experiment notebook. It covered score calculation, rolling and Mondrian calibration, band construction and projection, data preprocessing, masking, and walk-forward evaluation. Each surface is evaluated before its scores enter the calibration window, preserving temporal order. Chapter 5 reports the empirical comparison of Score A and Score B using the recorded coverage, width, projection, crossing, and prediction-error diagnostics.

Chapter 5

Experiments and Results

This chapter reports the empirical evaluation of the conformal uncertainty quantification framework developed in Chapter 3 and implemented in Chapter 4. The evaluation is organised around the six research objectives set out in Section 1.2: whether the framework achieves the nominal coverage level on held-out market data, how the rolling window behaves as recent surfaces enter and leave the calibration set, whether Mondrian binning addresses spatially uneven coverage, whether the projection reliably eliminates violations of the calendar and total-variance convexity conditions, and whether the resulting bands are sharp enough to be practically informative. Section 5.1 specifies the dataset and evaluation protocol. Section 5.2 examines quantile behaviour during the calibration phase. Section 5.3 presents the projection results for both scores and establishes why Score B is the recommended construction. Section 5.4 reports the full Score A versus Score B comparison across coverage, sharpness, and projection tractability. Sections 5.5 and 5.6 then report the detailed coverage and sharpness results for Score B as the primary framework. Section 5.7 summarises the findings.

5.1 Experimental Setup

The raw dataset comprises 56 S&P 500 (SPX) option surface files with collection dates between 11 December 2025 and 15 August 2026. The option chains were obtained through OpenBB using its Cboe provider and stored in a WRDS-compatible CSV schema, as described in Section 4.7. They are loaded by the `WRDSOptionsDataset` class, whose name refers to the expected schema rather than the source of the observations. Ten files carry collection dates on which no regular Cboe trading session occurred and are therefore excluded before the analysis dataset is constructed. The early-close session on 24 December 2025 is retained. This calendar filter leaves 46 surfaces spanning 11 December 2025 to 11 August 2026.

The first 15 retained surfaces, dated from 11 December 2025 to 17 February 2026, form the initial calibration period. The remaining 31 surfaces, dated from 3 March 2026 to 11 August 2026, are evaluated sequentially using a walk-forward procedure. At each evaluation step, the band is constructed from the preceding calibration window before the current surface is added, thereby preventing future observations from entering the calibration data. Missing dates are neither filled nor interpolated. Consequently, a rolling window of size $L = 5$ contains the five most recent available surfaces rather than five consecutive calendar days.

Within every retained surface, a fixed coordinate-stratified mask assigns 80% of the available quotes to the GNO context set and holds out 20% as target observations. The same deterministic masking rule, with seed 2026, is used for both score constructions. Coverage, prediction error, and band width are evaluated only at these held-out target locations. Each retained surface contains between 8,677 and 11,325 quotes in the transformed coordinate domain $D = (0.01, 1) \times (-1.5, 0.5)$, corresponding

to 6,943–9,058 context points and 1,734–2,267 held-out target points per surface.

The structure of the filtered dataset is illustrated in Figure 4.1 of Chapter 4, which shows the option quote count and the calibration/evaluation timeline across the 46 retained surfaces.

The point predictor underlying the framework is the pre-trained GNO of Gonon, Jacquier, and Wiedemann [10], loaded from the checkpoint `train/store/9448705/checkpoints/checkpoint_final.pt`. The model has 102,529 parameters and is run in inference mode without retraining or fine-tuning on the SPX data used here. The conformal layer is applied post-hoc, as described in Chapter 3: the GNO point predictions are fixed, and calibration and band construction are performed solely through the conformal pipeline. The configuration is fixed before walk-forward evaluation: nominal coverage $1 - \alpha = 90\%$, rolling-window size $L = 5$, 4×4 Mondrian binning, shrinkage $\lambda = 0.3$, and minimum bin population 10. These settings are listed in Table 4.2 of Chapter 4. The projection is solved by the joint linear program with fixed-k calendar constraints described in Section 3.5.2.

The primary evaluation is conducted using Score B (Vega-weighted implied volatility error, Section 3.2.2) with Vega-IV bands (Section 3.4.2), which is the construction recommended by this thesis. Score A (spread-normalised price error, Section 3.2.1) with price-spread bands (Section 3.4.1) is evaluated under identical conditions as an ablation, to examine how the choice of nonconformity score affects coverage across surfaces, sharpness, and the tractability of the finite-grid shape projection. Because both scores are applied to the same GNO predictions on the same dataset, any differences in the reported metrics are attributable solely to the nonconformity score and band construction.

5.2 Calibration Behaviour

During the calibration phase the pipeline processes each of the 15 calibration surfaces in chronological order. After each surface is added to the rolling window, the global and Mondrian quantiles are recomputed on the scores currently in the window. This section reports the behaviour of those quantiles for Score B across the calibration period, since Score B is the primary construction and its quantiles are the ones used for the main coverage evaluation in Section 5.5.

5.2.1 Rolling Window Warm-up

The rolling window accumulates surfaces until it reaches its capacity of $L = 5$. During this warm-up period, surfaces 0 through 4 (11 December 2025 to 2 January 2026), the calibration set grows with each step. The global 90% conformal quantile is 0.016392 after the first surface and 0.015963 once the window is full at surface 4. Each warm-up surface contributes between 1,734 and 1,799 held-out target scores. Thus, even the first step contains 1,792 calibration scores, while the full initial window contains 8,844 scores.

From surface 5 onward the window operates in steady state, discarding the oldest surface as each new one is added. The global quantile increases from 0.016275 at surface 5 to a calibration-period maximum of 0.018828 at surface 9 (15 January 2026), before declining to 0.016942 at surface 14 (17 February 2026). Across all 15 calibration steps it ranges from 0.015929 to 0.018828. These movements show that the rolling procedure updates the threshold as the five-surface score distribution changes.

5.2.2 Global versus Mondrian Quantiles

Throughout the calibration period the Mondrian quantiles span a substantially wider range than the single global quantile, indicating that Score B nonconformity varies across the implied volatility surface. At surface 0, when the window contains only one surface, the Mondrian bin quantiles range from

0.012486 to 0.042169, a ratio of approximately 3.4 between the lowest and highest bin. At surface 14, after the window has operated in steady state for ten steps, the range is $[0.011922, 0.039472]$, a ratio of approximately 3.3.

This spatial variation is the type of heterogeneity that Mondrian conformal prediction [19, 2] is designed to capture. The global construction applies one calibration threshold across the full domain, whereas the Mondrian construction assigns each point the quantile of the bin containing its (ρ, z) coordinates. Both constructions subsequently use the local Vega weight when converting the quantile into an implied volatility band. The practical effect of the bin-specific thresholds on evaluation coverage and sharpness is examined in Sections 5.5 and 5.6.

Table 5.1 reports the global and Mondrian quantile range at each of the 15 calibration surfaces for Score B, and Figure 5.1 displays the same information graphically alongside the per-surface mean Score B value.

Idx	Date	Global $\hat{q}$	Mondrian min	Mondrian max
0	2025-12-11	0.016392	0.012486	0.042169
1	2025-12-23	0.016425	0.012290	0.037176
2	2025-12-24	0.016201	0.012427	0.038016
3	2025-12-26	0.015929	0.012565	0.034061
4	2026-01-02	0.015963	0.012381	0.036271
5	2026-01-06	0.016275	0.012097	0.041657
6	2026-01-07	0.016555	0.012097	0.041741
7	2026-01-09	0.016706	0.011328	0.041786
8	2026-01-14	0.017996	0.011269	0.042173
9	2026-01-15	0.018828	0.011969	0.041741
10	2026-01-23	0.018144	0.012339	0.036615
11	2026-01-29	0.017855	0.012308	0.036528
12	2026-01-30	0.018111	0.012368	0.039823
13	2026-02-11	0.017336	0.012179	0.039590
14	2026-02-17	0.016942	0.011922	0.039472

Table 5.1: Score B calibration quantiles across 15 surfaces.

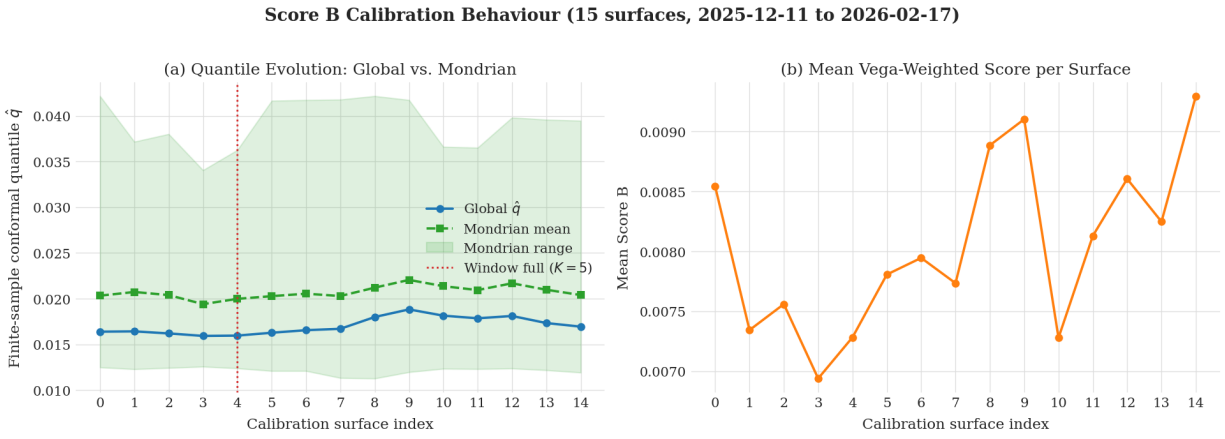


Figure 5.1: Score B global quantile $\hat{q}$, Mondrian quantile range, and per-surface mean score across the 15 calibration surfaces.

5.3 Finite-Grid Shape Projection

The conformal construction is complemented by a post-processing step that imposes two specified shape conditions on the lower and upper band surfaces. As discussed in Section 3.5.2, the routine operates on a rectangular (ρ, z) grid. The fixed-k calendar condition compares consecutive maturities at the same log-moneyness $k = \rho z$. The total-variance convexity condition compares each middle total-variance value with the linearly interpolated value between its two neighbouring z points. Both band surfaces are adjusted jointly so that their lower/upper ordering is included directly. This second condition does not enforce convexity of call prices in strike. The projection is applied before the final coverage and violation counts are evaluated.

5.3.1 Aggregate Violation Counts

For each evaluation surface, fixed-k calendar and total-variance convexity violations are counted separately on the lower and upper Mondrian bands and then summed. Table 5.2 reports the aggregate counts over all 31 walk-forward evaluation surfaces. It also reports quote-level band crossings after projection, defined as locations where the projected lower bound exceeds the projected upper bound.

Score	Before	After	Reduction	Crossings after
Score A (spread)	63,177	0	100.0%	0
Score B (Vega)	57,848	0	100.0%	0

Table 5.2: Combined fixed-k calendar and total-variance convexity violation counts on the lower and upper Mondrian bands across 31 walk-forward evaluation surfaces.

Figures 5.2 and 5.3 illustrate the enforced conditions on the lower Score B band for the first evaluation surface (surface 15, 3 March 2026). The left panels show selected slices before projection and the right panels show the corresponding slices after projection. Because the figures display selected lower-band slices, their violation counts differ from the aggregate totals in Table 5.2.

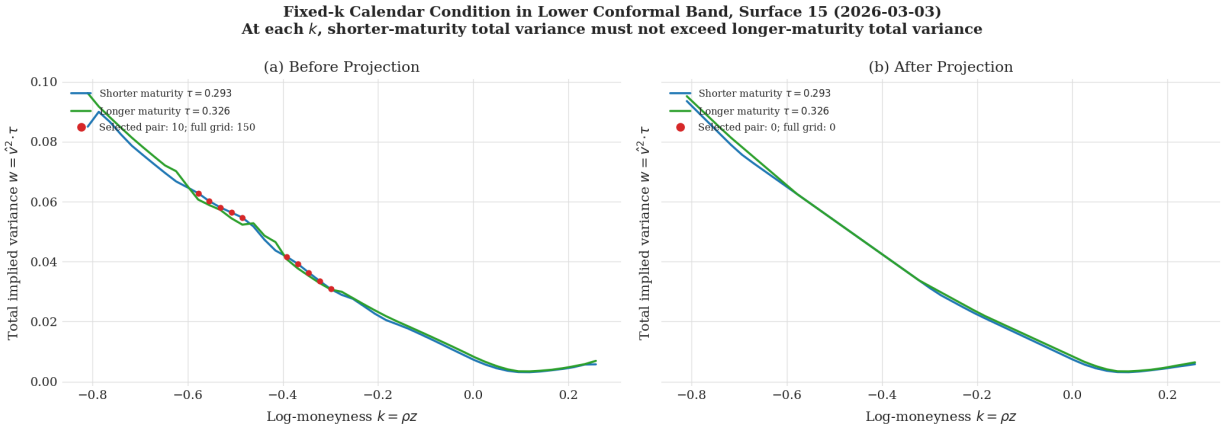


Figure 5.2: Calendar condition in selected slices of the lower Score B band for surface 15, before and after projection.

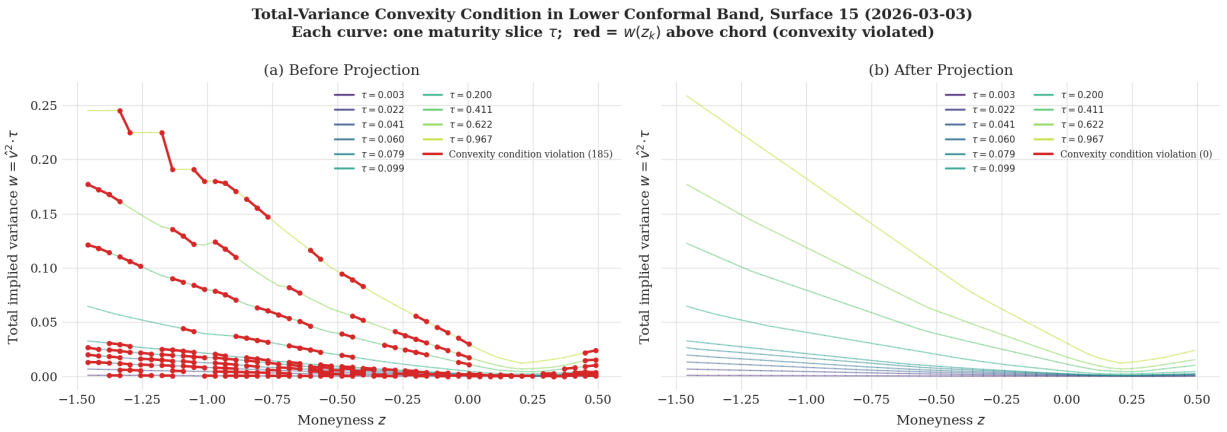


Figure 5.3: Total-variance convexity condition in selected maturity slices of the lower Score B band for surface 15, before and after projection.

5.3.2 Score B: Enforced Violations Eliminated

For Score B, the projection reaches zero residual violations of the two enforced grid conditions on every evaluation surface, reducing the combined lower-and-upper count from 57,848 to zero. The projected bands also have zero quote-level crossings across all 31 surfaces. This confirms that the joint linear program consistently enforces its fixed-k calendar, total-variance convexity, and lower–upper ordering constraints on the discretised surfaces; it does not establish butterfly no-arbitrage.

5.3.3 Score A: Enforced Violations Eliminated

For Score A, the same enforced violation count decreases from 63,177 to zero, and the projected bands have zero quote-level crossings across all 31 surfaces. The projection records 278 small positive-variance floor adjustments across both bounds for Score A, compared with none for Score B. Projection feasibility under the enforced conditions therefore does not by itself distinguish the two scores; their empirical coverage, sharpness, and numerical behaviour are compared in Section 5.4.

The supplementary GNO-style butterfly diagnostic is evaluated separately and is not imposed as a constraint by the linear program. Its aggregate violation count decreases from 25,946 to 7,432 for Score A and from 27,899 to 2,730 for Score B. These non-zero diagnostic counts mean that eliminating the enforced discrete violations should not be interpreted as establishing full absence of static arbitrage under every analytic condition.

5.4 Nonconformity Score Comparison

The projection results in Section 5.3 show that both scores reach zero residual violations of the enforced grid conditions. The two constructions nevertheless differ materially in their band width, empirical coverage after projection, and numerical behaviour. This section compares these outcomes to explain why Score B is selected as the primary construction.

5.4.1 Score Magnitude and Quantile Scale

The two scores use different normalisations and therefore operate at different numerical scales. Score B is a Vega-weighted IV residual. Because both experiments use the same GNO predictions, their mean

prediction error is identical at MAE $= 0.006813$ IV units. For Score B, the global calibration quantile is 0.016942 on the final calibration surface.

Score A is a spread-normalised price residual. On the first calibration surface (11 December 2025), its median is 1.1370, its 90th percentile is 4.0384, and its maximum is 23.3982. The corresponding global conformal quantile is 4.059961 and rises to values of approximately 20–23 as the rolling window advances, ending at 22.397682 on the final calibration surface. Score A divides the price residual by the bid–ask spread, subject to the implemented minimum spread of 0.10. Because the Score A and Score B quantiles enter different band-construction formulas, Equations 3.9 and 3.11, their raw magnitudes are not directly comparable. Their practical effects are instead compared below using IV-space band width and coverage.

5.4.2 Score A Inversion and Clipping Diagnostics

Score A first constructs price-space endpoints and then converts each endpoint to implied volatility. Table 5.3 reports the aggregate inversion diagnostics for the 65,791 held-out target quotes across the 31 evaluation surfaces. Of the lower price endpoints, 58,190 (88.45%) were inverted without adjustment. The remaining 7,601 lower endpoints (11.55%) were negative and were therefore clipped to the minimum IV of 10^{-4}. All upper endpoints were inverted successfully. There were no maximum-IV clips, numerical solver failures, or non-finite price endpoints for either bound.

Diagnostic	Lower endpoint	Upper endpoint
Total inversions	65,791	65,791
Successful inversions	58,190 (88.45%)	65,791 (100.00%)
Clipped to minimum IV	7,601 (11.55%)	0 (0.00%)
Clipped to maximum IV	0 (0.00%)	0 (0.00%)
Solver failures	0 (0.00%)	0 (0.00%)
Negative price endpoints	7,601 (11.55%)	0 (0.00%)
Non-finite price endpoints	0 (0.00%)	0 (0.00%)

Table 5.3: Aggregate Score A price-to-IV inversion diagnostics across the 31 evaluation surfaces.

The minimum bid–ask spread of 0.10 was applied to 1,667 of the 65,791 target quotes (2.53%). The diagnostics show that the numerical issue is concentrated in the lower price endpoint: more than one in ten lower endpoints fall below zero before inversion, while the upper endpoints require no clipping. This asymmetry is relevant when interpreting the post-projection coverage and width of Score A.

5.4.3 Band Width

Table 5.4 reports IV band-width summaries for both scores across the 31 evaluation surfaces. Each entry gives equal weight to each surface. The pre- and post-projection median rows give the means of the 31 within-surface median widths at the corresponding stage.

Metric	Score A (spread)	Score B (vega)
Mean width before projection (IV)	0.060798	0.027118
Mean width after projection (IV)	0.067806	0.027187
Average of per-surface median widths before projection (IV)	0.024051	0.025277
Average of per-surface median widths after projection (IV)	0.033759	0.025850
Mean of per-surface percentage width changes	9.88%	0.254%

Table 5.4: Surface-weighted IV band-width summaries across the 31 evaluation surfaces.

After projection, Score A is approximately 2.5 times wider than Score B in mean IV width and 1.3

times wider in the mean of the per-surface median widths. The mean of the 31 per-surface percentage width changes from the direct to the projected bands is 9.88% for Score A and 0.254% for Score B. This averages the percentage change on each surface. Calculating a percentage from the two reported mean widths instead gives $(0.067806 - 0.060798)/0.060798 \approx 11.53\%$ for Score A. The two summaries differ because the pre-projection width varies across surfaces. Score B therefore provides sharper final bands and has a smaller width change across the combined grid mapping and optimisation procedure.

5.4.4 Coverage Across Evaluation Surfaces

Table 5.5 reports the post-projection per-surface Mondrian coverage for both scores alongside the overall mean and standard deviation across all 31 evaluation surfaces. The summary statistics give equal weight to each surface; they are not calculated by pooling all quotes across dates.

Idx	Date	Score A coverage	Score B coverage
15	2026-03-03	91.31%	91.42%
16	2026-04-10	73.81%	91.60%
17	2026-04-16	82.98%	94.36%
18	2026-04-17	79.36%	89.34%
19	2026-05-08	82.05%	96.50%
20	2026-05-13	82.16%	91.24%
21	2026-05-14	86.38%	95.58%
22	2026-05-15	83.71%	94.45%
23	2026-05-18	80.06%	94.58%
24	2026-05-19	92.32%	91.81%
25	2026-05-20	92.39%	94.47%
26	2026-05-21	96.17%	94.93%
27	2026-05-22	85.13%	94.81%
28	2026-06-04	82.18%	89.32%
29	2026-06-05	72.13%	91.36%
30	2026-06-16	81.77%	93.51%
31	2026-06-23	85.17%	93.41%
32	2026-06-24	94.29%	93.60%
33	2026-06-26	84.83%	94.02%
34	2026-07-02	87.74%	93.14%
35	2026-07-07	82.02%	91.25%
36	2026-07-08	76.25%	92.24%
37	2026-07-15	85.53%	95.45%
38	2026-07-17	86.28%	95.88%
39	2026-07-30	94.67%	91.04%
40	2026-07-31	81.43%	92.02%
41	2026-08-04	86.88%	93.01%
42	2026-08-05	89.88%	89.03%
43	2026-08-06	77.72%	90.96%
44	2026-08-07	72.80%	92.02%
45	2026-08-11	82.81%	92.42%
Mean		**84.27%**	**92.86%**
Std		**6.24%**	**1.99%**

Table 5.5: Post-projection per-surface Mondrian coverage for Score A and Score B across all 31 evaluation surfaces. Target: $1 - \alpha = 90\%$.

Figure 5.4 shows the post-projection per-surface coverage for both scores, together with the 90% nominal target. Figure 5.5 separately shows their mean IV band widths across the same 31 evaluation surfaces.

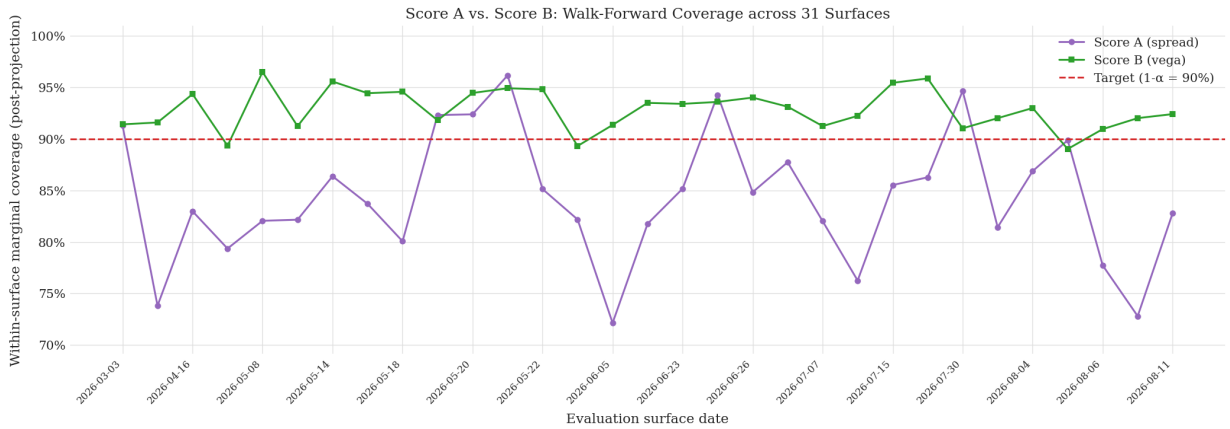


Figure 5.4: Post-projection per-surface Mondrian coverage for Score A and Score B across the 31 evaluation surfaces.

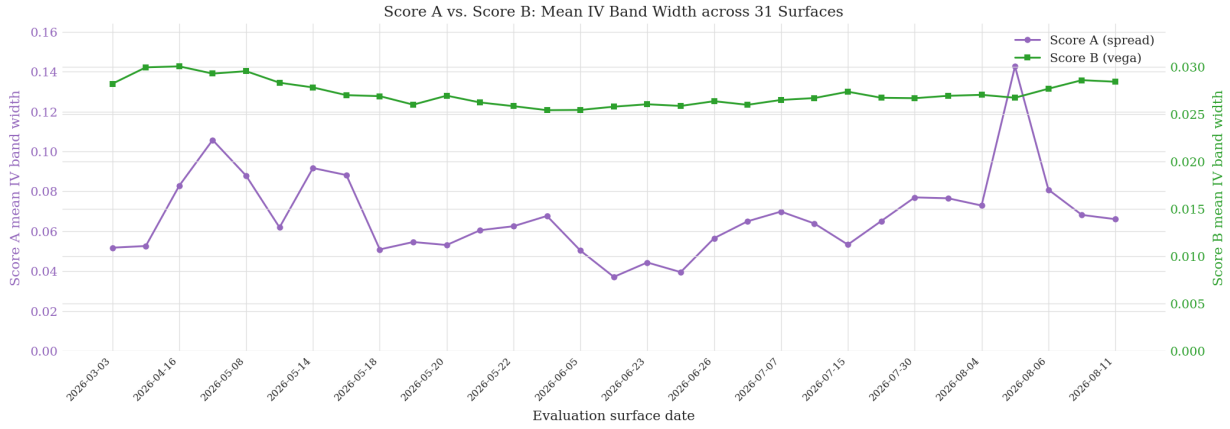


Figure 5.5: Post-projection mean IV band width for Score A and Score B across the 31 evaluation surfaces. The two vertical axes show the different width scales of the two scores.

Before projection, mean Mondrian coverage is 90.19% for Score A and 91.12% for Score B. After projection, Score A coverage decreases to 84.27%, whereas Score B coverage increases to 92.86%. The surface-bootstrap 95% confidence interval for the post-projection mean is $[82.10\%, 86.42\%]$ for Score A and $[92.20\%, 93.54\%]$ for Score B.

Post-projection Score A coverage ranges from 72.13% to 96.17% and falls below the nominal target on 25 of the 31 surfaces. Its standard deviation is 6.24 percentage points. Score B ranges from 89.03% to 96.50%, falls below the target on three surfaces, and has a standard deviation of 1.99 percentage points. Score B therefore provides both higher mean coverage and substantially greater temporal stability after projection.

5.4.5 Separating Interpolation and Optimisation Effects

The direct-to-projected comparison combines the quote-to-grid and grid-to-quote interpolation with the intervening optimisation. To distinguish them, the unadjusted grid bands are also mapped back to the held-out target quotes without solving the optimisation problem, using the same interpolation and original-endpoint fallback as the projected bands. Table 5.6 compares these three stages at the same 65,791 target quotes. Coverage, width, and GNO-prediction containment are first calculated within

each surface and then averaged across the 31 surfaces with equal weight. The outside counts pool quotes across surfaces.

Score	Stage	Coverage	Mean width (IV)	GNO inside	GNO outside
A (spread)	Direct	90.19%	0.060798	100.00%	0
A (spread)	Interpolation only	91.45%	0.060932	99.90%	65
A (spread)	Projected	84.27%	0.067806	91.03%	5,945
B (vega)	Direct	91.12%	0.027118	100.00%	0
B (vega)	Interpolation only	92.06%	0.027170	99.93%	43
B (vega)	Projected	92.86%	0.027187	99.85%	95

Table 5.6: Matched Mondrian-band coverage, width, and GNO-prediction containment at the direct, interpolation-only, and projected stages. Percentage and width entries are equally weighted across surfaces; outside counts are pooled across quotes.

Interpolation alone raises Score A mean coverage by 1.26 percentage points, from 90.19% to 91.45%, with a mean width increase of 0.000133 IV. The subsequent optimisation lowers coverage by 7.18 percentage points to 84.27%, while increasing mean width by a further 0.006875 IV. At this second transition, 6,116 previously covered market quotes leave their intervals and 1,349 previously uncovered quotes enter. The same transition removes the GNO prediction from 5,907 intervals and brings it into 27. After optimisation, 5,945 of the 65,791 intervals do not contain the GNO prediction, compared with 65 after interpolation alone. For Score B, the interpolation-only and projected coverages are 92.06% and 92.86%, respectively; projected GNO-prediction containment remains 99.85% on an equal-surface basis.

Table 5.7 reports how each step moves the lower and upper endpoints at the held-out quotes. Signed means show the average direction, while means of absolute movements retain local adjustments that cancel in a signed average. In particular, Score A's optimisation changes the lower endpoint by 0.028729 IV in mean absolute value, much more than its upper endpoint (0.001284 IV). Its signed mean lower shift is negative, yet the large absolute movement and the quote-level losses show that the wider average band does not imply coverage at each location.

Score	Transition	Δlo	$\lvert\Delta\text{lo}\rvert$	Δhi	$\lvert\Delta\text{hi}\rvert$
A (spread)	Interpolation	-0.000050	0.004538	$+0.000083$	0.001382
A (spread)	Optimisation	-0.007164	0.028729	-0.000289	0.001284
B (vega)	Interpolation	$+0.000024$	0.000746	$+0.000076$	0.000754
B (vega)	Optimisation	-0.000117	0.000995	-0.000100	0.000869

Table 5.7: Mean signed and absolute endpoint movements in IV units, equally weighted across evaluation surfaces. Interpolation compares direct with interpolation-only bands; optimisation compares interpolation-only with projected bands.

5.4.6 Summary of Score Comparison

Both scores reach zero violations of the enforced calendar and total-variance convexity conditions, so that result alone does not determine the preferred construction. For Score A, interpolation alone improves coverage; the observed coverage loss occurs during the optimisation of its grid bands. The comparison between scores therefore describes the complete score, band-construction, and post-processing pipelines, rather than isolating a failure of the Score A nonconformity score. Score B remains narrow, changes little in width, and achieves a post-projection mean coverage of 92.86% with lower variation across surfaces. Accordingly, Sections 5.5 and 5.6 report Score B as the recommended construction.

5.5 Coverage Results

This section examines the empirical marginal coverage of the Score B framework in detail across the 31 held-out evaluation surfaces. Coverage is the fraction of market implied volatility quotes that fall within the constructed band at each evaluation surface, as defined in Section 3.6. Two variants are reported at the same post-projection stage: the global construction uses a single scalar quantile from the rolling window, whereas the Mondrian construction uses per-bin quantiles from the same window. Both use the same $L = 5$ rolling window and the same projection routine; the difference is the quantile assignment.

5.5.1 Per-Surface Coverage

Table 5.8 reports the matched post-projection global and Mondrian coverage for Score B on each evaluation surface. The mean and standard deviation give equal weight to each of the 31 surface-level coverage values.

Idx	Date	Global	Mondrian	Idx	Date	Global	Mondrian
15	2026-03-03	88.12%	91.42%	31	2026-06-23	90.41%	93.41%
16	2026-04-10	91.17%	91.60%	32	2026-06-24	90.05%	93.60%
17	2026-04-16	91.20%	94.36%	33	2026-06-26	90.23%	94.02%
18	2026-04-17	89.39%	89.34%	34	2026-07-02	90.72%	93.14%
19	2026-05-08	93.72%	96.50%	35	2026-07-07	88.01%	91.25%
20	2026-05-13	89.61%	91.24%	36	2026-07-08	89.26%	92.24%
21	2026-05-14	92.73%	95.58%	37	2026-07-15	93.64%	95.45%
22	2026-05-15	92.17%	94.45%	38	2026-07-17	94.57%	95.88%
23	2026-05-18	92.65%	94.58%	39	2026-07-30	90.14%	91.04%
24	2026-05-19	89.59%	91.81%	40	2026-07-31	91.59%	92.02%
25	2026-05-20	90.87%	94.47%	41	2026-08-04	89.19%	93.01%
26	2026-05-21	90.50%	94.93%	42	2026-08-05	88.98%	89.03%
27	2026-05-22	93.39%	94.81%	43	2026-08-06	90.47%	90.96%
28	2026-06-04	90.11%	89.32%	44	2026-08-07	92.42%	92.02%
29	2026-06-05	89.69%	91.36%	45	2026-08-11	89.52%	92.42%
30	2026-06-16	92.02%	93.51%				
Mean		**90.84%**	**92.86%**				
Std		**1.68%**	**1.99%**				

Table 5.8: Matched post-projection Score B coverage on all 31 evaluation surfaces ($1 - \alpha = 90\%$).

Over the full 31-surface evaluation period, the mean Mondrian coverage is 92.86%, above the 90% nominal target. Its standard deviation is 1.99 percentage points, and the surface-bootstrap 95% confidence interval for its mean is $[92.20\%, 93.54\%]$. The minimum Mondrian coverage is 89.03% (surface 42, 5 August 2026), and the maximum is 96.50% (surface 19, 8 May 2026). Mondrian coverage falls below the 90% target on three evaluation surfaces.

5.5.2 Contribution of Mondrian Spatial Binning

The global coverage column of Table 5.8 provides the comparison without spatial binning while retaining the same rolling window and projection stage. Its mean coverage is 90.84%, compared with 92.86% for Mondrian coverage.

Mondrian spatial binning raises mean post-projection coverage by 2.02 percentage points. For example, coverage increases from 88.12% to 91.42% on surface 15 and from 89.19% to 93.01% on surface 41. Across the evaluation period, eight surfaces have global coverage below 90% but Mondrian coverage

at or above 90%. These results indicate that assigning calibration quantiles by surface region reduces undercoverage on several evaluation dates. The cross-surface standard deviation is 1.68 percentage points for global coverage and 1.99 percentage points for Mondrian coverage, so the improvement in mean coverage is not accompanied by lower temporal variation.

The three surfaces where Mondrian coverage remains below 90% are surfaces 18, 28, and 42. Global coverage reaches at least 90% only on surface 28, showing that Mondrian binning improves average coverage but does not increase coverage on every individual surface. At each walk-forward step, coverage is evaluated using the preceding rolling window; the current surface is added only after its evaluation and can then contribute to the next window. The window therefore contains the five most recently processed surfaces without using future information. The remaining temporal variation and its relation to the conformal assumptions are discussed further in Chapter 6.

Figure 5.6 compares post-projection global and Mondrian coverage within each cell of the 4×4 partition. Coverage is first calculated within each populated bin on every evaluation surface and then averaged across surfaces, so that dates with more target quotes do not receive greater weight.

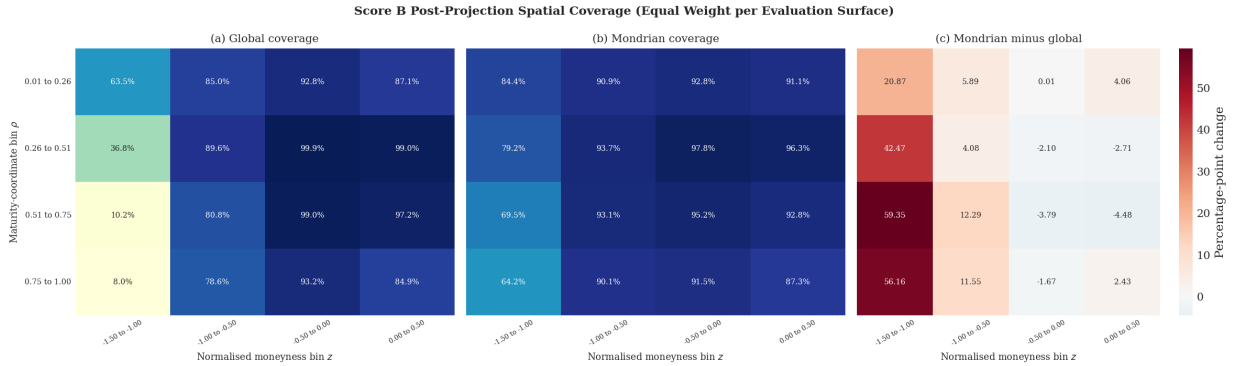


Figure 5.6: Surface-weighted post-projection Score B coverage by (ρ, z) bin for global and Mondrian calibration. The right panel reports the Mondrian-minus-global change in percentage points.

Table 5.9 reports the number of held-out evaluation quotes in each bin, pooled across the 31 surfaces, and the post-projection Score B widths under global and Mondrian calibration. Each width is calculated within a bin on each surface and then averaged with equal weight across surfaces. All 16 bins are populated on all 31 surfaces; their target counts sum to 65,791.

ρ bin	z bin	Targets	Global width	Mondrian width
$[0.0100, 0.2575)$	$[-1.50, -1.00)$	1,639	0.03294	0.05235
$[0.0100, 0.2575)$	$[-1.00, -0.50)$	3,734	0.03301	0.04049
$[0.0100, 0.2575)$	$[-0.50, 0.00)$	9,010	0.03295	0.03297
$[0.0100, 0.2575)$	$[0.00, 0.50)$	5,624	0.03301	0.03785
$[0.2575, 0.5050)$	$[-1.50, -1.00)$	1,085	0.03298	0.05587
$[0.2575, 0.5050)$	$[-1.00, -0.50)$	3,474	0.03290	0.03459
$[0.2575, 0.5050)$	$[-0.50, 0.00)$	10,507	0.03011	0.02206
$[0.2575, 0.5050)$	$[0.00, 0.50)$	5,716	0.02923	0.02299
$[0.5050, 0.7525)$	$[-1.50, -1.00)$	469	0.03297	0.06592
$[0.5050, 0.7525)$	$[-1.00, -0.50)$	1,658	0.03293	0.04027
$[0.5050, 0.7525)$	$[-0.50, 0.00)$	7,716	0.02183	0.01558
$[0.5050, 0.7525)$	$[0.00, 0.50)$	3,214	0.01979	0.01616
$[0.7525, 1.0000)$	$[-1.50, -1.00)$	279	0.03295	0.06287
$[0.7525, 1.0000)$	$[-1.00, -0.50)$	1,366	0.03295	0.03986
$[0.7525, 1.0000)$	$[-0.50, 0.00)$	7,318	0.01663	0.01484
$[0.7525, 1.0000)$	$[0.00, 0.50)$	2,982	0.01438	0.01541

Table 5.9: Held-out target populations and surface-weighted mean post-projection IV band widths by (ρ, z) bin for Score B.

The largest coverage increases in Figure 5.6 occur in the lowest-moneyness column, $z \in [-1.5, -1.0]$, where the change ranges from 20.87 to 59.35 percentage points across the four maturity bins. Table 5.9 shows that this column contains only 279 to 1,639 held-out targets per bin and that its Mondrian widths are 0.05235–0.06592 IV units, compared with global widths of 0.03294–0.03298. Despite the wider bands, Mondrian coverage in this column remains between 64.2% and 84.4%, below the nominal target in every maturity bin. In the remaining cells, coverage changes range from -4.48 to 12.29 percentage points. Mondrian calibration therefore reduces severe spatial undercoverage in the lowest-moneyness region, but it does not eliminate it and does not improve coverage in every bin.

5.6 Sharpness

Sharpness is assessed through the width of the prediction bands at the held-out target quotes. Coverage alone does not distinguish a narrow band from an unnecessarily wide one, so band width is reported alongside coverage. The results below concern the post-projection Mondrian bands on the 31 evaluation surfaces.

5.6.1 Overall Band Width

Averaging the per-surface mean IV band widths across the 31 evaluation surfaces, with equal weight given to each surface, gives 0.027187 IV units. The surface-bootstrap 95% confidence interval for this mean is $[0.026758, 0.027644]$. The corresponding equal-weight average of the per-surface median widths is 0.025850 IV units. These values are computed at the scattered target quotes after the projected grid values have been mapped back to the original (ρ, z) coordinates.

Before projection, the mean Score B width is 0.027118 IV units. Projection raises it by 0.000069 IV units on average, corresponding to an increase of 0.254%. The GNO point-prediction MAE is 0.006813 IV units. The post-projection mean half-width of 0.013594 IV units is therefore approximately twice the MAE. Together with the mean Mondrian coverage of 92.86%, these values summarise the observed coverage and band width of Score B.

5.6.2 Spatial Variation of Band Width

The Mondrian construction permits spatially varying band widths by assigning calibration quantiles according to the (ρ, z) bin containing each point. On surface 14, the last surface in the initial calibration period, the bin quantiles range from 0.011922 to 0.039472 (Table 5.1). The largest threshold is approximately 3.3 times the smallest. This comparison concerns the calibration thresholds, not the final band widths, which also depend on the local Vega weight and the projection. The post-projection global and Mondrian widths by bin are reported in Table 5.9. As shown in Section 5.5.2, the spatial assignment increases mean post-projection coverage relative to the global construction, although it does not increase coverage on every surface.

Figure 5.7 illustrates the post-projection Score B bands for the first evaluation surface, dated 3 March 2026. The left panel shows the GNO estimate together with the lower and upper conformal surfaces. The right panel shows the pointwise distance between the two conformal surfaces in IV units.

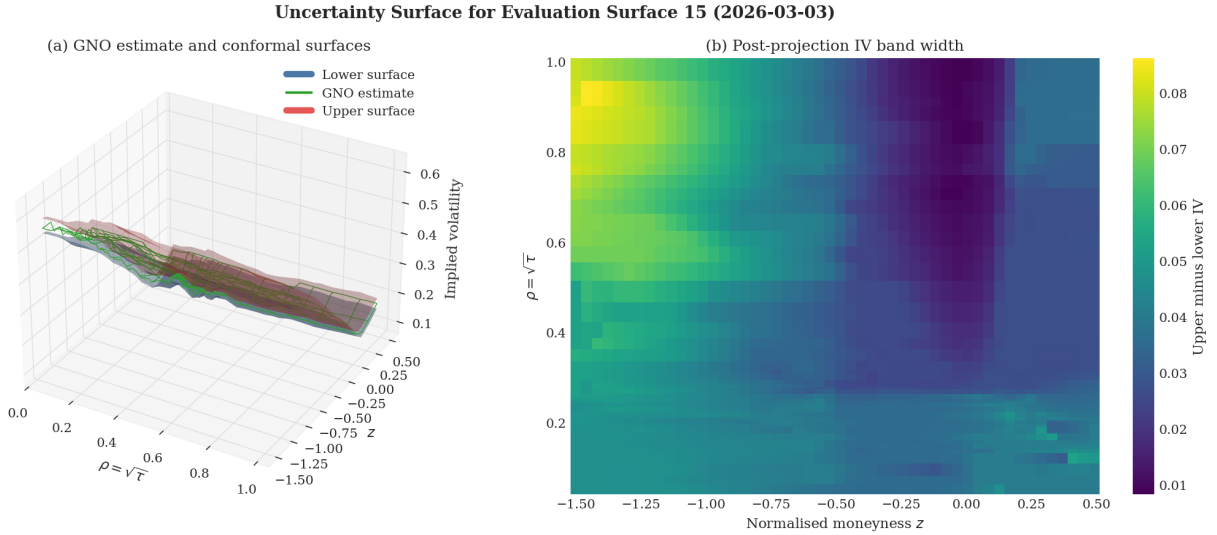


Figure 5.7: GNO implied-volatility estimate, post-projection lower and upper Score B conformal surfaces, and the corresponding IV band width for evaluation surface 15.

For this surface, the largest widths occur mainly at negative normalised moneyness and longer maturities. The smallest widths appear near the centre of the moneyness range at longer maturities. This figure provides one spatial example; the aggregate width and coverage results use all 31 evaluation surfaces.

5.6.3 Sharpness Relative to Score A

The comparison in Section 5.4.3 uses the same 31 evaluation surfaces and the same post-projection stage for both scores. Score A has a mean band width of 0.067806 IV units, approximately 2.49 times the Score B mean of 0.027187. Its mean Mondrian coverage is 84.27%, compared with 92.86% for Score B. In this experiment, Score B therefore has both the narrower mean band and the higher post-projection coverage. The difference reflects the score constructions: Score A calibrates spread-normalised price residuals and maps price bands into IV space, whereas Score B forms the bands from Vega-weighted IV residuals.

5.7 Summary

The evaluation began with 56 collected S&P 500 option surface files. After excluding ten files whose collection dates did not correspond to regular trading sessions, 46 surfaces remained. The first 15 formed the initial calibration period, and the remaining 31 were processed in walk-forward order with a rolling window of size $L = 5$.

During the initial calibration period, the Score B global quantile ranged from 0.015929 to 0.018828. On surface 14, the final surface in this period, the Mondrian bin quantiles ranged from 0.011922 to 0.039472. The largest bin threshold was therefore approximately 3.3 times the smallest, reflecting spatial differences in the calibrated scores.

Across the 31 evaluation surfaces, Score B achieved mean post-projection Mondrian coverage of 92.86%, with a surface-bootstrap 95% confidence interval of $[92.20\%, 93.54\%]$. The corresponding global coverage was 90.84%. Mondrian coverage was below the 90% target on three surfaces. The mean post-projection Score B width was 0.027187 IV units, with a 95% confidence interval of $[0.026758, 0.027644]$, while the GNO point-prediction MAE was 0.006813 IV units. Projection increased the mean Score B width by 0.254%.

The joint projection eliminated all violations of the enforced calendar and total-variance convexity conditions for both scores. The combined count fell from 63,177 to zero for Score A and from 57,848 to zero for Score B, with no lower and upper band crossings at the target quotes. The supplementary GNO-style butterfly diagnostic was not imposed as a constraint and remained nonzero after projection: its aggregate count decreased from 25,946 to 7,432 for Score A and from 27,899 to 2,730 for Score B.

The comparison of the two scores used the same predictions and evaluation surfaces. Score A had mean post-projection Mondrian coverage of 84.27%, a coverage standard deviation of 6.24 percentage points, and a mean width of 0.067806 IV units. For Score B, the corresponding values were 92.86%, 1.99 percentage points, and 0.027187 IV units. Score A was approximately 2.49 times wider on average while having lower coverage. Score B is therefore retained as the primary construction for the discussion in Chapter 6, where the temporal variation in coverage, the conformal assumptions, and the limitations of the projection diagnostics are examined.

Chapter 6

Discussion

This chapter interprets the empirical findings of Chapter 5 in broader context. Section 6.1 discusses the exchangeability assumption underlying conformal coverage and its relevance to the temporally ordered SPX options data. Section 6.2 considers the practical interpretation of the resulting band surfaces. Section 6.3 discusses the limitations of the current study, and Section 6.4 outlines focused directions for future research. Section 6.5 summarises the main points.

6.1 The Exchangeability Assumption in Practice

The standard finite-sample coverage guarantee of conformal prediction relies on exchangeability of the calibration and test nonconformity scores, meaning that their joint distribution is invariant to permutations of their indices [19, 2]. Independent and identically distributed observations provide a common sufficient condition, but exchangeability is the more general requirement. In the present application, option surfaces are ordered in time and the quotes within a surface are related through the same market state. Exact exchangeability is therefore not assumed. The rolling procedure is used as a practical adaptation mechanism, and its coverage is evaluated empirically in Chapter 5.

6.1.1 Serial Dependence and the Rolling Window

The rolling window contains the $L = 5$ most recently processed surfaces. Because the available dates are irregular, it represents the latest five observations rather than a fixed calendar span. Older surfaces leave the window as new surfaces are processed. This restriction does not establish exchangeability or restore the formal finite-sample guarantee under temporal dependence. Its purpose is to make the conformal quantile responsive to recent errors, consistent with the motivation for giving recent observations greater relevance under distribution drift [3].

During the initial calibration period, the Score B global quantile is 0.016392 after the first surface and 0.016942 after surface 14. Across the 15 calibration steps, it ranges from 0.015929 to 0.018828. These values document how the threshold changes as surfaces enter and leave the window. They do not establish exchangeability or rule out distributional change; they only describe the behaviour of the selected rolling procedure before the walk-forward evaluation.

6.1.2 Temporal Variation in Coverage

Across the 31 evaluation surfaces, post-projection global coverage is below the 90% target on ten surfaces, while Mondrian coverage is below 90% on three. The three Mondrian shortfalls occur on surfaces 18, 28, and 42, corresponding to 17 April, 4 June, and 5 August 2026. For example, on

surface 15 global coverage is 88.12% and Mondrian coverage is 91.42%, whereas surface 42 remains below the target at 88.98% globally and 89.03% with Mondrian assignment. The remaining shortfalls are therefore separated across the evaluation period rather than concentrated in one consecutive block.

At each walk-forward step, the current surface is evaluated before its scores are added to the window. The procedure can therefore respond only after a new error pattern has been observed; it cannot anticipate a change that is absent from the preceding surfaces. The available data do not include a separate analysis of market regimes or other covariates that would identify the cause of each coverage shortfall. Barber et al. [3] analyse conformal coverage under departures from exchangeability. Accordingly, the reported coverage is interpreted as an empirical result, and the standard exchangeable-data guarantee is not claimed for this time-dependent evaluation.

6.1.3 Contribution of Mondrian Binning

Mondrian binning raises mean post-projection coverage from 90.84% for the global construction to 92.86%, an increase of 2.02 percentage points. On eight surfaces where global coverage is below 90%, the Mondrian construction raises coverage to at least the target. It does not raise coverage on every surface, and surfaces 18, 28, and 42 remain below 90%. The cross-surface standard deviation is 1.68 percentage points for global coverage and 1.99 percentage points for Mondrian coverage. The higher mean coverage therefore does not correspond to lower temporal variation.

At surface 14, the bin-specific Score B quantiles range from 0.011922 to 0.039472 around the global quantile of 0.016942. Each target point receives a quantile according to its (ρ, z) region, while its final band width also depends on the local Vega weight and the finite-grid shape projection. Mondrian conformal prediction is motivated by calibration within predefined categories under exchangeability [20, 19]. Here, the observed quantile pattern is evidence that the implementation adapts spatially; it is not a claim of exact conditional coverage.

6.2 Practical Implications

The framework produces lower and upper surfaces around the smooth GNO estimate over the modelled domain. Calibration and empirical coverage are evaluated at held-out target locations drawn from observed market quotes. Their implied volatilities are withheld from the GNO during prediction and used afterwards to calculate nonconformity scores and coverage. Interpolation and projection then convert the quote-level bands into surface objects. The output can therefore be examined across the domain without replacing the quote-level coverage calculation.

For Score B, the post-projection Mondrian bands have a mean IV width of 0.027187 and a mean half-width of 0.013594. The GNO mean absolute error on the same held-out target quotes is 0.006813. Band width and prediction error measure different quantities: the former is determined by the recent calibration scores, local Vega weighting, and projection, while the latter measures point-prediction accuracy. Wider regions indicate greater empirically calibrated prediction uncertainty around the GNO estimate. The bands summarise recent out-of-sample prediction errors under the selected score, rolling calibration, and projection; they are not confidence intervals for the GNO parameters or direct trading or hedging recommendations.

The comparison favours Score B for this experiment. After projection, its mean Mondrian coverage is 92.86% with a mean IV width of 0.027187. The corresponding Score A coverage is 84.27%, with a mean width of 0.067806, approximately 2.5 times the Score B width. Score A also has greater variation across evaluation surfaces: the standard deviation of post-projection Mondrian coverage is 6.24 percentage points, compared with 1.99 percentage points for Score B. Score B is therefore retained as the primary construction because it exceeds the nominal coverage level with narrower bands and less

variation across surfaces in this evaluation. This comparison concerns the complete band-construction and post-processing pipelines, not the nonconformity scores in isolation.

The stage-wise comparison in Section 5.4.5 locates Score A's coverage deterioration in the optimisation stage: interpolation alone raises mean coverage from 90.19% to 91.45%, after which optimisation reduces it to 84.27%. The latter stage widens the band on average by 0.006875 IV but reduces GNO-prediction containment from 99.90% to 91.03%. The mean absolute lower-endpoint movement of 0.028729 IV shows that substantial local repositioning is hidden by the mean width. Thus, the observed loss cannot be attributed to the Score A score alone; it reflects how its bands respond to this finite-grid optimisation. Across both stages combined, mean width rises by 0.007008 IV for Score A and 0.000069 IV for Score B. For both scores, the projection eliminates all violations of the enforced calendar and total-variance convexity conditions on the evaluation grids. No lower-to-upper crossings are observed at the evaluated quotes. These finite-grid results apply to the conditions enforced by the implementation and do not establish absence of arbitrage under every possible diagnostic.

6.3 Limitations

The empirical study uses one underlying asset, one pre-trained GNO checkpoint, and 56 stored SPX option surface snapshots. Ten snapshots with non-regular-session dates are excluded, leaving 46 retained surfaces. The first 15 surfaces, from 11 December 2025 to 17 February 2026, form the initial calibration period, and the remaining 31 surfaces, from 3 March to 11 August 2026, form the walk-forward evaluation. The saved date is the collection date recorded by the retrieval script rather than a separate exchange-provided quote timestamp. Date-only collection and expiry fields determine the computed maturities; the unavailable intraday quote and settlement times limit their precision, particularly for short-dated contracts. The observations are irregular and do not form a complete sample of trading days. The findings are therefore specific to this dataset, model, and period.

A development-only sensitivity check was conducted for Score B before projection. Surfaces 0 to 6 provide the initial history, while surfaces 7 to 14 form a common validation period of eight surfaces. One parameter is varied at a time: $L \in \{3, 5, 7\}$, a 3×3, 4×4, or 5×5 Mondrian partition, and $\lambda \in \{0.1, 0.3, 0.5\}$. The 31 final evaluation surfaces are not used. Across the seven configurations, mean coverage ranges from 89.77% to 91.41%, and mean IV width ranges from 0.02794 to 0.02870. The reference configuration $L = 5$, 4×4, and $\lambda = 0.3$ gives 91.13% coverage and a mean width of 0.02832. Because the comparison uses one fixed context-target mask, only eight validation surfaces, and one-at-a-time parameter changes, it is a local sensitivity check rather than an exhaustive robustness or model-selection study.

Coverage is measured over the held-out target quotes within each evaluation surface and then averaged across surfaces. Quotes within a surface are cross-sectionally related, and regions containing more target quotes contribute more observations to that surface's coverage. The reported confidence intervals resample complete surfaces independently and do not model serial dependence between dates. Coverage and confidence intervals should therefore be interpreted as empirical summaries of the evaluation sample, not as exact finite-sample guarantees for the time series.

The projection enforces a fixed-k calendar condition and a total-variance convexity condition based on neighbouring grid points on a finite (ρ, z) grid. Their aggregate violation counts fall to zero after projection for both scores. Total-variance convexity is not equivalent to convexity of call prices in strike. A separate finite-difference evaluation of the GNO-style butterfly functional is retained as a supplementary diagnostic rather than an enforced constraint. Its post-projection counts remain nonzero: 7,432 for Score A and 2,730 for Score B across both bounds and all evaluation surfaces. The zero counts for the enforced conditions therefore do not establish butterfly no-arbitrage or that the bands are arbitrage-free over the continuous domain. Grid resolution and interpolation can also affect the

reported counts.

The lower and upper surfaces are adjusted jointly, with their ordering included in the linear optimisation on the projection grid. No crossings are observed after interpolation at the evaluated quote locations for either score. This check does not examine every point of the continuous surface between the grid nodes and observed quotes.

6.4 Future Research

Future empirical work should evaluate the framework on a longer sequence of SPX surfaces, on additional index and single-name option datasets, and with more than one trained GNO checkpoint. Such experiments would test whether the relative performance of Score A and Score B, the observed band widths, and the effect of projection persist across market periods, underlyings, and model fits.

Temporal adaptation should be studied using market-state variables or predefined volatility regimes. Weighted or adaptive conformal methods could then be compared with the fixed rolling window when recent observations differ from older calibration data [3]. The comparison should retain the walk-forward order so that information from a surface is unavailable until after that surface has been evaluated.

Parameter robustness requires a longer validation period, repeated context-target masks, and joint changes to the window length, Mondrian grid, and shrinkage parameter. Model selection and final evaluation should remain separate. Confidence intervals could also use a block or time-series resampling method that represents dependence between neighbouring dates rather than resampling surfaces independently.

For the shape adjustment, the supplementary GNO-style butterfly functional could be incorporated directly into the projection objective or constraints and compared with the total-variance convexity condition currently enforced. Repeating the analysis on finer grids would help identify violations that occur between the current grid nodes. Comparisons with other uncertainty constructions should report coverage, band width, computational cost, enforced-condition counts, and the separate GNO-style butterfly diagnostic under the same evaluation masks.

6.5 Summary

The walk-forward experiment contains 31 evaluation surfaces. For Score B, mean post-projection coverage increases from 90.84% with global calibration to 92.86% with Mondrian calibration, and the post-projection Mondrian bands have a mean IV width of 0.027187. The corresponding Score A construction has mean coverage of 84.27% and a mean width of 0.067806. Its cross-surface coverage standard deviation is also higher, at 6.24 percentage points compared with 1.99 percentage points for Score B. Within this evaluation, Score B has higher coverage, narrower bands, and lower variation across surfaces.

For both scores, projection eliminates all violations of the enforced calendar and total-variance convexity conditions on the evaluation grids, and no lower-to-upper crossings are observed at the target quotes. The supplementary GNO-style butterfly diagnostic remains nonzero after projection, with aggregate counts of 7,432 for Score A and 2,730 for Score B. The conclusions are consequently limited to the implemented finite-grid conditions and the observed SPX sample. Temporal dependence, the use of one GNO checkpoint, and the small development-only sensitivity check also prevent the empirical results from being interpreted as universal coverage or arbitrage guarantees.

Chapter 7

Conclusion

7.1 Summary of the Study

This thesis addressed the absence of calibrated uncertainty information around implied volatility surfaces produced by a deterministic Graph Neural Operator. The pre-trained GNO provides a smooth estimate from an irregular set of option quotes, but it returns a point prediction rather than a range describing the uncertainty around that estimate. The contribution of this work is a post-hoc conformal framework that constructs lower and upper surfaces without changing the GNO architecture or retraining its parameters.

The framework combines the main methodological components examined throughout the thesis. A coordinate-stratified mask separates each retained option surface into context observations and held-out targets for calibration and evaluation. Score A measures a price residual relative to the observed bid–ask spread, while Score B applies a Vega-based weight to the implied volatility residual. Exact conformal order statistics are computed from a rolling window, and Mondrian binning provides spatially varying thresholds. The resulting bands can be represented in both price and implied volatility space. Finally, the lower and upper surfaces are adjusted jointly to enforce the calendar condition, the total-variance convexity condition based on neighbouring grid points, positivity, and lower–upper ordering on the projection grid.

The empirical study used a curated sequence of stored SPX option surfaces. Files collected outside regular trading sessions were excluded, and the retained surfaces were divided into an initial calibration period and a chronological walk-forward evaluation. At each step, only preceding surfaces were available for calibration; the current surface entered the window after its evaluation. Score A and Score B therefore used the same GNO predictions, context-target masks, and evaluation sequence.

7.2 Main Findings

The rolling calibration behaved as intended: its threshold changed as surfaces entered and left the window. The Mondrian thresholds also varied across the option domain, showing that the implementation responds to both recent prediction errors and spatial differences in those errors.

For Score B, Mondrian calibration achieved higher mean coverage than the global construction and exceeded the nominal target on average. It also corrected several surface-level shortfalls, although it did not improve every evaluation surface. The lowest-moneyness region remained the most difficult part of the domain, showing that spatial calibration reduced but did not remove regional differences in coverage.

The Score B bands remained comparatively narrow, and the projection changed their average width

only slightly. The observed coverage was therefore not obtained through a large expansion of the bands during the financial-shape adjustment.

The matched comparison between the two nonconformity scores favoured Score B. The spread-normalised price construction produced wider bands, lower coverage, and greater variation between surfaces. The Vega-weighted IV construction therefore provided the stronger combination of empirical coverage, sharpness, and stability in this experiment.

The projection achieved its stated finite-grid objective for both scores by eliminating all observed violations of the enforced calendar and total-variance convexity conditions. No lower–upper crossings were observed at the evaluated target quotes after projection. A separate GNO-style butterfly diagnostic, which was not imposed as a constraint, still detected violations. The result therefore shows that the implemented discrete conditions were enforced successfully, but it does not establish that every static-arbitrage condition holds over the continuous domain.

7.3 Overall Conclusion

The research objectives were met at the level of implementation and empirical evaluation. Two economically motivated nonconformity scores were constructed, rolling calibration and Mondrian spatial binning were integrated with a pre-trained GNO, bands were obtained in price and implied volatility space, and the lower and upper surfaces were subjected to joint calendar and total-variance convexity adjustments. The walk-forward results show that the choice of score is consequential. For the observed SPX sample, Score B is the preferred construction because it achieved mean coverage above the nominal level with narrower bands and less variation across surfaces than Score A. Mondrian calibration improved average coverage relative to one global threshold, although the remaining regional and temporal shortfalls show that spatial binning does not remove every source of miscalibration.

These findings are limited to one underlying, one GNO checkpoint, a relatively short and irregular evaluation sequence, and one fixed context-target mask. The rolling window makes calibration responsive to recent observations but does not restore the exchangeability required for the standard conformal guarantee. The sensitivity analysis used a short development period and one-at-a-time parameter changes, while the projection conclusions apply only to the implemented grid conditions. A broader assessment should use longer and more varied datasets, multiple trained checkpoints and masks, adaptive or weighted conformal methods, time-aware resampling, and finer projection grids. The supplementary GNO-style butterfly diagnostic could also be incorporated directly into the projection and compared with the current discrete condition.

Within these limits, the study shows that conformal prediction can be added to a neural-operator volatility smoother as a practical uncertainty layer without retraining the underlying model. The resulting surfaces retain the computational advantage of a fixed GNO while attaching empirically evaluated ranges to its point estimates. On the available sample, Vega-weighted conformal calibration produced informative spatially varying bands, coverage above the nominal target on average, and only a small change in width from the financial-shape projection.

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